Integrated Design and Reproducible Feasibility Analysis of a Four-Wheel Rover with Packet-Commanded Imaging, LiDAR Ranging, and Learning-Assisted Target Recognition

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Abstract— This study develops an integrated, reproducible design assessment for a four-wheel educational rover intended for the high-school remote-control division of the 2027 NASA Human Exploration Rover Challenge. The architecture combines commercial differential drive control, independently sprung wheel carriers, separate drive and task batteries, and a Raspberry Pi-based subsystem that executes received imaging and ranging commands. The analysis reconciles component lists with dimensioned computer-aided design views, derives mobility and electrical constraints, and connects sensor geometry to command execution. At the estimated mass of 10.919 kg, a 215 mm wheel rotating at 200 RPM produces a theoretical rolling speed of 8.105 km/h; consequently, the nominal motor label does not establish compliance with an 8 km/h design ceiling. A steady 20° climb requires at least 1.255 N · m per wheel under an assumed rolling-resistance coefficient of 0.10 and equal torque sharing. At 50 ft, an ideal 12-inch feature spans approximately 71 pixels across a full-resolution Camera Module 3 Standard image, while a 1° pointing error displaces the ranging ray by approximately 266 mm. An executed synthetic benchmark compares color–geometry rules and two linear support-vector classifiers on 2,400 generated board crops with scene-disjoint evaluation. A separate reference command executor is checked using mocked hardware. These software results and analytical sensitivities support a testable development plan; the study does not report physical rover trials. The principal contribution is a traceable connection between component choices, failure mechanisms, quantitative design limits, and reproducible verification tasks.

Index Terms— Educational robotics, skid steering, independent suspension, command autonomy, camera–LiDAR integration, synthetic data, support-vector machines, reproducible engineering.

1. Introduction

A small rover that can cross uneven terrain and execute a radio-commanded scientific task presents an integration problem rather than a collection of independent component-selection problems. Wheel

diameter determines the relationship between motor speed and vehicle speed. Suspension travel changes load distribution and sensor orientation. Motor transients can disturb computing and radio reception. A plausible range value may refer to a background surface rather than the intended target. These interactions matter particularly when the same vehicle must meet a competition envelope, operate from batteries, and present its results immediately after an excursion.

The 2027 NASA Human Exploration Rover Challenge (HERC) remote-control high-school rules provide the application context [1]. The rover may be driven into position, after which the specified task operations are autonomous. In this paper, *autonomy* means executing received task commands without manual camera triggering, filter selection, or range measurement. It does not mean autonomous navigation around the entire course. This distinction permits a commercial radio-controlled drivetrain to coexist with a separately programmed task subsystem and prevents the capability statement from exceeding the proposed implementation.

The mechanical basis is a student-supplied four-wheel design with a sloped equipment enclosure, custom printed wheels, and independent pivoting wheel carriers. The electrical basis comprises four brushed geared motors, two electronic speed controllers (ESCs), and separate lithium-polymer batteries for propulsion and task electronics. The task system combines a Raspberry Pi 5, a camera, a single-beam LiDAR, a servo-positioned sensor assembly, an onboard display, and a receive-only software-defined radio chain. The supplied material includes CAD images and planned component lists, but no native SolidWorks assembly, calibrated mass properties, motor performance curves, or physical test logs. The research question is therefore: **which parts of the proposed architecture are quantitatively consistent, which limits can be derived, and which measurements are necessary before the design can be regarded as demonstrated?**

Three contributions address that question. First, the study reconciles the latest dimensional and mass estimates with mobility, suspension, power, and optical calculations. Second, it specifies an executable command-state architecture and a hardware-independent verification suite covering command order, requests for fresh captures, fault handling, and logical state preservation. Third, it evaluates a learning-assisted target-classification extension on a reproducible synthetic dataset, with explicit separation between nominal and shifted scene distributions. The resulting manuscript is a design and software feasibility study; its experiments concern generated images and mocked interfaces rather than a completed rover.

2. Evidence Basis, Related Work, and Design Requirements

2.1. Literature retrieval and scope

A broad discovery search conducted on 15 September 2026 used 20 OpenAlex queries spanning skid steering, rover suspension, wheel–soil interaction, printed PETG/TPU structures, camera–range calibration, and synthetic vision. The queries returned 4,000 metadata occurrences. Removing 547 duplicates left 3,453 unique records; automated term matching in titles and available abstracts retained 1,514 topic candidates and excluded 1,939. This is a documented metadata screen, not a systematic review or a claim that thousands of full papers were read. Twenty-five primary research references were selected for closer relevance assessment: 14 received selective full-text inspection, while 11 were assessed at the available abstract or metadata level. The accompanying evidence ledger records these access levels and the limits of each claim.

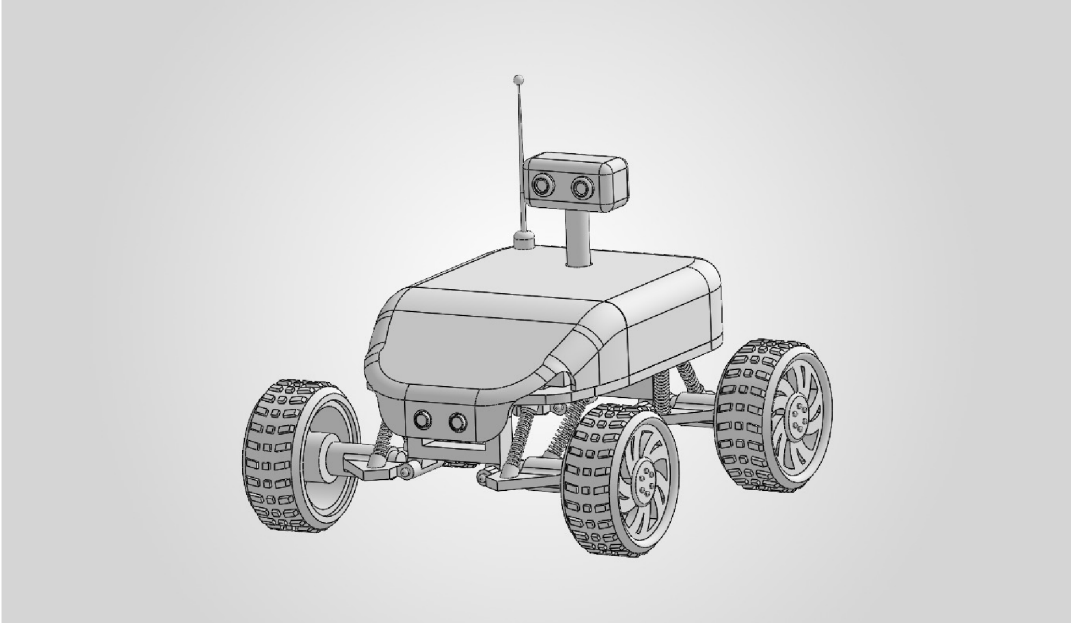
The search is useful for identifying established failure mechanisms and suitable analysis methods. It does not justify importing another rover's friction coefficients, suspension benefits, classification accuracy, or fatigue life. Search ranking, absent abstracts, and broad keywords also prevent a defensible claim of exhaustive coverage. Manufacturer documents and the supplied handbook are treated separately from the research-paper screen because they establish interface specifications and task rules rather than independent experimental evidence.

2.2. Mobility, suspension, and printed wheels

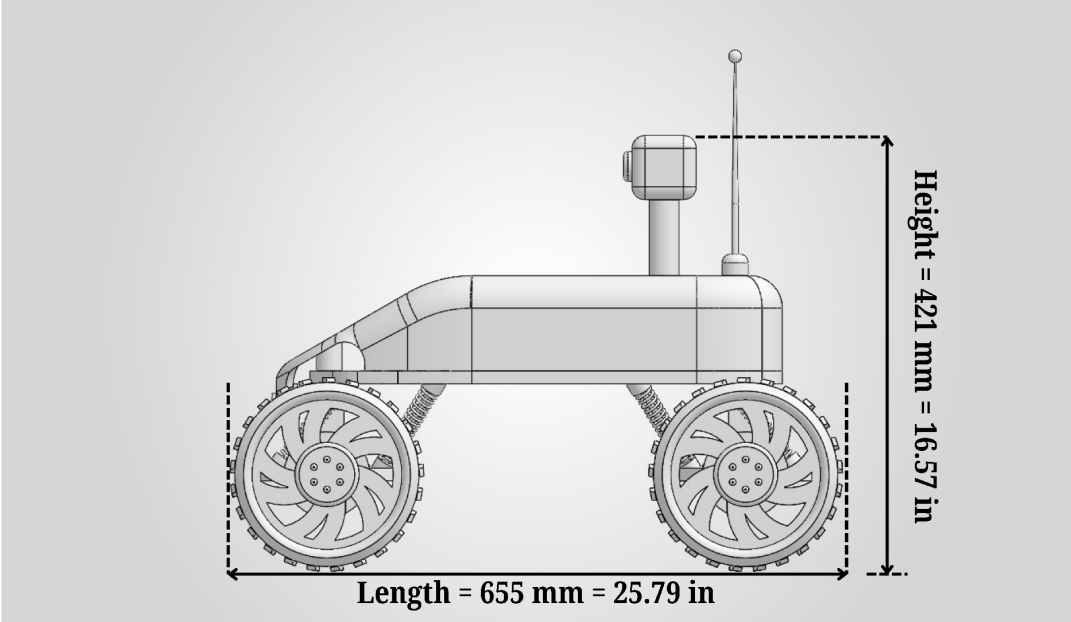
Skid steering requires lateral slip during a turn. Mandow et al. identify effective kinematic parameters from measurements, while Wang et al. study curvature-dependent instantaneous centers of rotation on a particular four-wheel platform [2, 3]. These studies support using ideal differential-drive equations as

Supplied CAD: assembly and side dimensions

A | Three-quarter assembly concept



B | Length 655 mm; height to camera 421 mm



Design illustrations; the 421 mm height excludes the antenna.

Figure 1: Supplied rover CAD views: assembly and side profile. These are design renderings supplied by the team, not photographs of a fabricated prototype. The latest dimensioned design takes precedence over earlier concept images.

an initial command model, followed by terrain-specific identification. They do not establish universal turning parameters for the present rover. Dynamic modeling also requires rolling resistance, terrain shear, actuator saturation, and controller behavior [4].

Terramechanics studies provide a basis for investigating wheel sinkage and traction [5–7]. However, material response, soil preparation, wheel geometry, and load are necessary inputs. Sutoh et al. demonstrate that diameter, mass, and grouser configuration can alter climbing behavior in their loose-soil tests and that classical models do not necessarily predict lightweight-rover performance precisely [8]. For this reason, the present calculations use transparent resistance sweeps rather than inventing a calibrated soil model.

The supplied suspension is also distinct from the adaptive rocker concept discussed by Reina and Foglia [9]. It uses independent pivot arms and paired springs, rather than a rocker-type mechanism with variable wheel-camber compensation. Work on spring-suspended skid-steer rovers motivates considering suspension-dependent wheel contact geometry [10], while the Mars Exploration Rover mobility literature illustrates the separate importance of design and physical verification [11]. Neither source makes the current CAD assembly a flight-qualified or terrain-validated mechanism.

Printed-wheel literature supports a component-level characterization program. Wang et al. manufactured TPU specimens and non-pneumatic tires and reported sensitivity to manufacturing conditions [12]. PETG studies similarly show that printed response depends on processing parameters and reinforcement [13, 14]. Competition-rover wheels with separately printed rims and compliant tires provide a fabrication precedent [15]. More recent PETG/TPU multi-material work and PLA/TPU topology studies broaden the design space [16, 17], but their material systems and measured properties cannot be assigned directly to the proposed unfilled PETG rim and TPU 95A tire.

2.3. Sensing and learning

Camera calibration establishes the mapping between pixels and viewing rays. Zhang's planar-target method provides an established foundation for identifying camera intrinsics and distortion [18]. Camera–range registration is a separate problem: approaches using multiple checkerboards or alignment of camera and laser features demonstrate the importance of overlapping geometry and independent calibration [19, 20]. Those methods were developed for richer range observations than the single beam available here. The present design therefore requires an instrumented, multi-pose calibration procedure adapted to a single-beam sensor.

Synthetic-data studies motivate controlled variation of lighting, texture, pose, and background [21–23]. They do not imply that a classifier trained on simple generated boards will transfer to outdoor camera images. Dynamics randomization concerns physical/control variation [24], while domain adaptation can combine simulated and real visual experience [25]. These are distinct mechanisms. Terrain classification using proprioceptive measurements is another relevant rover application of machine learning [26], but the present experiment is restricted to visual classification of a supplied board crop.

2.4. Traceable design basis

The latest technical specification is used where earlier drafts conflict. In particular, the camera-height dimension is 421 mm, the rim diameter is 188 mm, and the tire radial thickness is 13.5 mm. Earlier values of 432 mm, 185 mm, and 15 mm are not mixed into the calculations. The 421 mm height excludes the antenna; a complete excursion-ready envelope still requires a measurement that includes every installed projection.

Table 1: Adopted design quantities and their evidence status.

Quantity	Adopted value or interpretation	Evidence status
Overall length × width	655 × 630 mm	Supplied design dimensions
Height to camera top	421 mm, antenna excluded	Supplied design dimension
Wheels	Four; 215 mm diameter × 90 mm width	Supplied design dimensions
Rim / tire radial thickness	188 mm / 13.5 mm	Dimension closure verified
Estimated fully equipped mass	10.919 kg	Team-supplied theoretical estimate; unweighed

Quantity	Adopted value or interpretation	Evidence status
Track / wheelbase	540 / 440 mm	Conditional inference from outer dimensions
Motors	Four 12 V brushed geared units, nominally about 200 RPM	Listed class; continuous curves unavailable
Drive / task battery	3S, 5,000 / 2,200 mAh	Planned components
Task computer / camera	Raspberry Pi 5, 8 GB / Camera Module 3 Standard	Planned components
Range sensor	TF03-100 with serial interface	Planned component
Camera pan mechanism	Servo-positioned, bearing-supported platform	Proposed; usable angular range unverified

The inferred track assumes that the stated width is the outer tire width and that each tire is symmetrically located: $630 - 90 = 540$ mm. The inferred wheelbase assumes that the front and rear wheel extrema establish overall length: $655 - 215 = 440$ mm. These are conditional contact-geometry estimates, not extracted native-CAD measurements.

Table 2: Handbook requirements mapped to verification needs.

Handbook consideration	Design interpretation	Evidence needed before acceptance
Excursion-ready cube, Rule 5.5.4	762 mm on each axis	Measure complete rover, including antenna
High-school manufacture, Rule 6.4.2.1	Team-designed/manufactured chassis, wheels and tires; permitted commercial drive/suspension parts	Manufacturing records and final configuration
Primary drive control, Rule 6.4.3	Commercial transmitter, receiver and ESC control	Demonstration of configured control and failsafe
Speed, Rule 6.4.4	Use 8 km/h as the conservative ceiling associated with the stated 5 mph limit	Full-charge maximum-speed test and limiter verification
Mass, Rules 6.4.5 and Section 8.2	60 lb absolute limit; 20 lb is a scoring objective	Weigh excursion-ready vehicle
Battery protection, Rules 6.4.6–6.4.7	Separate insulated enclosures; main overcurrent protection	Wiring and enclosure inspection
Task autonomy, Rule 6.4.8	Manual positioning followed by automatic task execution	Complete received-command sequence demonstration

These interpretations refer to the supplied 2027 handbook [1]. The calculated mass is 24.072 lb, below the stated 60 lb absolute limit but above the 20 lb scoring objective. Reaching 20 lb would require removing 1.847 kg, or 16.92% of the present estimate. The calculation is a mass-management target, not a reason to remove load-bearing material without re-evaluating strength.

3. Integrated Mechanical and Drive Architecture

3.1. Chassis, wheel carriers, and load paths

The proposed chassis uses 6061-T6 aluminum tube and sheet to support the equipment enclosure, batteries, suspension pivots, and sensor mast. Bolted connections provide assembly access and allow individual modules to be replaced. The battery masses should be placed low and secured against movement, because a loose pack changes both electrical reliability and center of mass. The enclosure must permit cooling airflow and service access without interfering with wheel travel or exposed moving joints.

Each wheel carrier pivots relative to the chassis and carries its motor, spindle supports, wheel, and spring attachments. Keeping the motor and spindle on the same moving carrier preserves their relative alignment as the suspension articulates. A coupling connects the nominal 6 mm motor shaft to the wheel spindle. The wheel spindle is supported by two separated bearings and positively retained at the hub. This load path is intended to transfer wheel contact forces through the spindle bearings and carrier rather than assuming that a small gearbox output bearing can sustain terrain impacts.

The CAD concept shows two coils at each wheel, with one damper associated with one of the coils. Thus the complete rover has eight springs and four dampers. The architecture should not be described as having two dampers per wheel. Before fabrication, the final assembly must establish pivot-axis alignment, bearing fits, end stops, spring seating, cable bending radii, and full-travel interference clearances. Screenshots cannot reveal interference through every degree of freedom or substitute for an assembly-motion study.

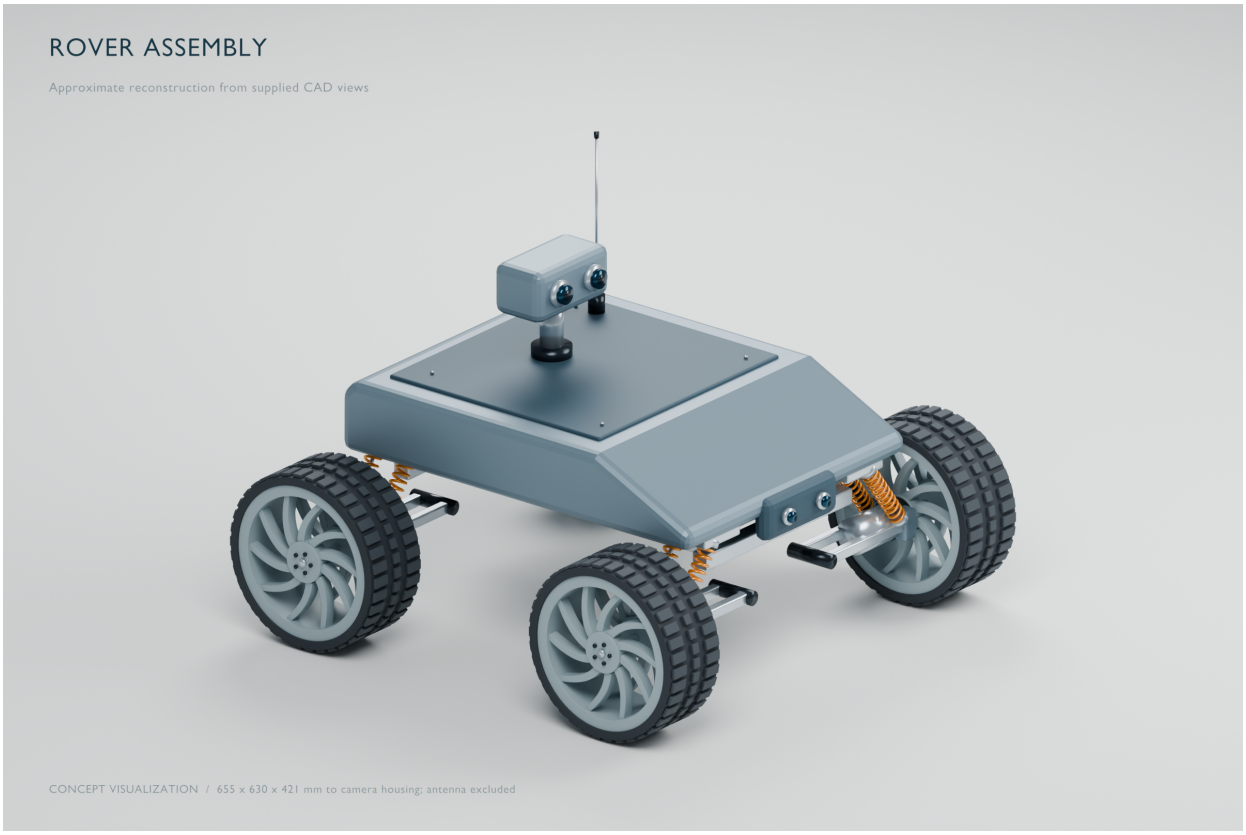


Figure 2: New Blender reconstruction of the proposed rover. Geometry is an approximate reconstruction from the supplied CAD views and dimensions; the rendering illustrates packaging and appearance, not a manufactured vehicle or structural simulation.

3.2. Wheel manufacture and structural verification

The outer wheel dimension closes geometrically as

$$D_o = D_r + 2t = 188 + 2(13.5) = 215 \text{ mm}.$$

The proposed PETG rim provides a relatively stiff hub and spoke structure, while the TPU 95A tire supplies a compliant tread. "95A" identifies a nominal hardness class; it does not establish the stiffness, fatigue resistance, or tear strength of the printed tire. The manufacturing record should therefore include filament identity, drying procedure, extrusion temperature, layer height, wall count, infill topology and fraction, print orientation, and any bonding or mechanical interlock between rim and tire. The printed interface requires its own retention check.

A defensible strength analysis begins with final geometry and measured or appropriately conservative material properties. For a spindle idealized as a solid, circular, linearly elastic shaft under bending and torsion, bending and torsion give

$$\sigma_b = \frac{32M}{\pi d^3}, \qquad \tau_t = \frac{16T}{\pi d^3}, \qquad \sigma_{\text{vm}} = \sqrt{\sigma_b^2 + 3\tau_t^2}.$$

These expressions require the bearing span, overhang, bending moment, applied torque, and actual spindle diameter. None of those missing inputs is inferred from the visual thickness of a rendered part.

Bearing reactions must be computed from the real support locations; a hub overhang can increase bearing reactions beyond the simple wheel-load value. Thread roots, keyways, printed notches, and joint preload need local treatment rather than a smooth-shaft formula alone.

Printed coupons should first characterize the selected process in the relevant directions. Rim/hub tests should then apply radial, lateral, and torsional loads representative of the assembly. Tire tests should measure load–deflection behavior, permanent set, traction on the intended surfaces, and tread or interface damage after repeated use. A finite-element model can subsequently be correlated to these measurements. No stress contour or safety factor is presented here because the required native geometry, boundary conditions, and material characterization were not supplied.



Figure 3: Published FDM TPU tire used as a manufacturing comparison. Reproduced from J. Wang et al., original Fig. 12 [12], under CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). This is the cited authors' tire, not a photograph of the present rover wheel; its properties are not transferred to this design.

3.3. Commercial differential drive and electrical connections

The transmitter mixes throttle and steering into two centered command channels. The receiver sends one pulse-width control signal to the left ESC and one to the right ESC, with their required signal returns. Each reversible brushed ESC drives the two motors on its side in parallel. Forward and reverse operation therefore change the polarity across the motor pair; neither motor output should be treated as a permanent chassis ground. Motor direction is checked with the wheels raised before the rover is allowed to move.

Only one ESC battery-eliminator circuit (BEC) supplies the receiver. The positive BEC conductor from the other ESC is isolated while its signal and reference return remain correctly connected. This avoids assuming that two independent regulated outputs can be paralleled. Receiver failsafe and transmitter-loss behavior must command neutral on both sides, and the test must include loss of radio reception under motor load. Neutral pulse values and ESC calibration are specific to the selected hardware.

The drive pack feeds an accessible disconnect and a proposed 20 A main fuse. A proposed 5 A time-delay fuse is included in one lead of each individual motor branch. These numbers are provisional coordination choices. Their sum is not a calculation of operating current, and a label rating does not establish interruption capacity, transient tolerance, wire protection, or selectivity. Motor-start, turn, stall, and sustained-climb currents must be measured with the final wiring and load.

For normalized throttle u and steering s, an initial mixer is

$$a = u - s, \quad b = u + s, \quad q = \max(1, |a|, |b|), \quad u_L = a/q, \quad u_R = b/q.$$

This jointly normalized mixer is a conceptual allocation rule. Its implementation must be checked against the commercial transmitter's actual mixing and saturation behavior; independent channel clipping does not preserve the same command ratio. Joint normalization retains the left/right ratio while respecting the allowed command range. Sign conventions must be verified against the installed motor orientation. This is a mixing relationship, not a calibrated mapping from stick position to ground speed.

Ideal differential-drive kinematics give

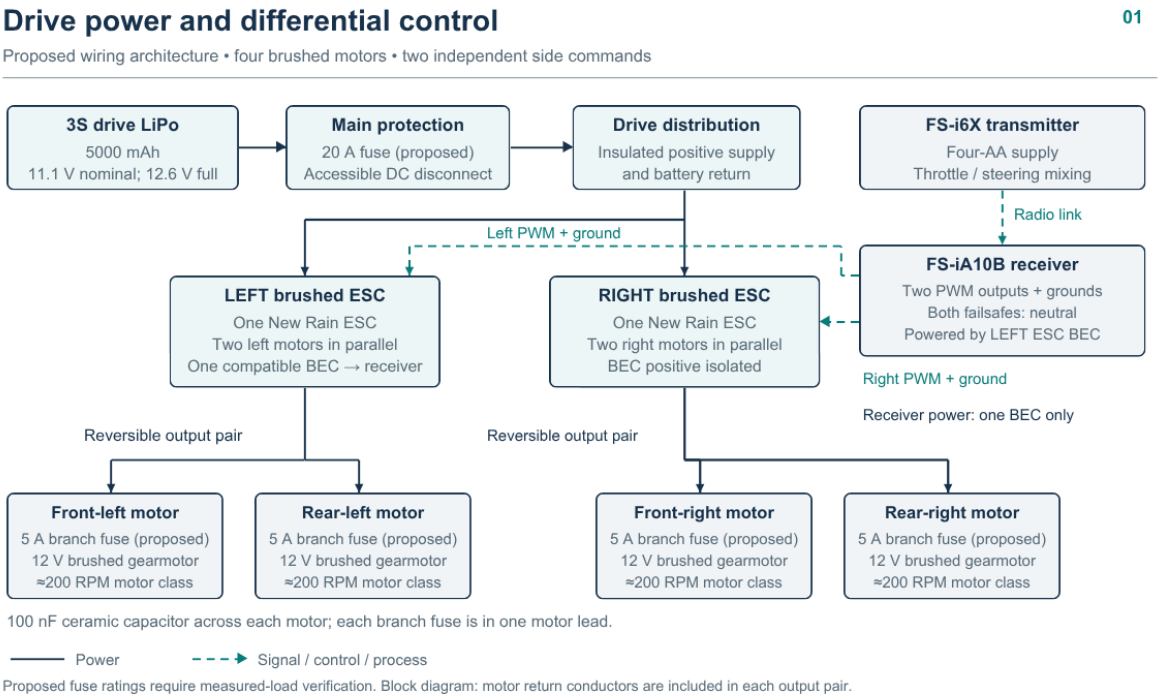


Figure 4: Editable Canva drive-system block diagram. It separates receiver commands, the single active BEC supply, battery distribution, and reversible motor outputs. Each motor-output connection represents a conductor pair; detailed connector pinouts remain a wiring-drawing task.

$$v_x = \frac{r}{2}(\omega_R + \omega_L), \qquad \dot{\psi} = \frac{r}{T_{\text{eff}}}(\omega_R - \omega_L).$$

Here ω_L and ω_R are wheel angular speeds in rad/s. The effective separation T_{eff} equals geometric track only under the idealized model. In actual skid steering it depends on slip, terrain, curvature, wheel loading, and suspension configuration [2, 3]. Changing effective separation models yaw sensitivity but does not itself compensate longitudinal slip. A pivot turn is therefore a demanding current and traction case, not a "zero-radius" capability established by the wiring diagram. The proposed drivetrain does not include a demonstrated encoder-based autonomous velocity controller.

4. Mobility, Traction, and Stability Analysis

4.1. Wheel speed and the full-charge case

For wheel diameter D in meters and rotational speed n in RPM, ideal rolling speed is

$$v_{\text{km/h}} = \frac{\pi D n\, 60}{1000}.$$

At $D = 0.215$ m, 190 RPM gives 7.70004 km/h and 200 RPM gives 8.10531 km/h. The 8 km/h boundary is

$$n_8 = \frac{8(1000)}{\pi(0.215)(60)} = 197.40148 \text{ RPM}.$$

Consequently, a nominal "200 RPM" motor description does not demonstrate compliance. The earlier assumption that loading will reduce speed to 190 RPM is a scenario, not a measurement. A light-load, high-voltage condition may be the relevant maximum-speed case.

If, and only if, a motor actually runs at 200 RPM at 12 V with approximately linear no-load voltage scaling, a fully charged 3S pack at 12.6 V would imply 210 RPM and 8.51057 km/h. The ratio 8/8.51057 is approximately 0.940. This is a first-pass endpoint estimate, not a verified ESC limiter setting: PWM

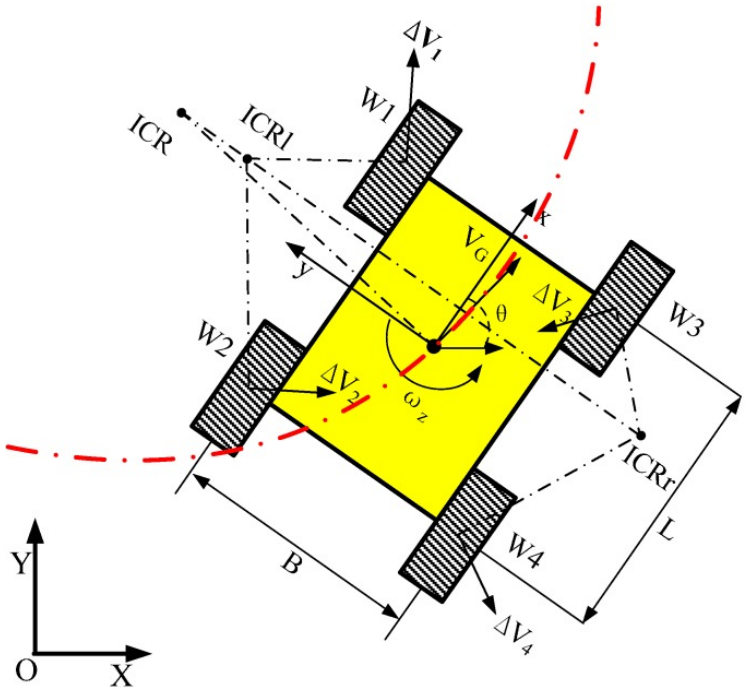


Figure 5: Published skid-steer instantaneous-center geometry, reproduced from T. Wang et al., original Fig. 1 [3], under CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). The symbols and geometry belong to that paper. The figure motivates measured slip-aware kinematics; no fitted coefficient is copied into the present rover model.

behavior, no-load specifications, tire deformation, battery sag, and terrain all affect the result. A lower provisional target, followed by a measured full-charge test over a known distance, provides room for calibration uncertainty. The limiter must be assessed in every permitted operating mode, including reverse if relevant.

4.2. Straight-climb lower bounds

Let m denote total mass, g gravitational acceleration, θ slope angle, and C_{rr} an assumed rolling-resistance coefficient. A steady straight climb with no acceleration or additional soil deformation is approximated by

$$F_{\text{req}} = mg(\sin\theta + C_{rr}\cos\theta), \qquad \tau_w = \frac{rF_{\text{req}}}{4}.$$

The equal-sharing expression gives the required output torque at each wheel. Motor-shaft torque before a gearbox is a different quantity. With available tire–terrain friction coefficient μ, the corresponding traction condition is

$$\mu mg\cos\theta \geq F_{\text{req}}, \qquad \mu \geq \tan\theta + C_{rr}.$$

This is a necessary global traction condition, not a sufficient per-wheel condition. Grade-dependent load transfer and unequal contact loads can cause an individual wheel to slip even when the aggregate inequality is satisfied.

For $m = 10.919$ kg, $r = 0.1075$ m, $g = 9.80665$ m/s^2, and $C_{rr} = 0.10$, the resulting values are as follows.

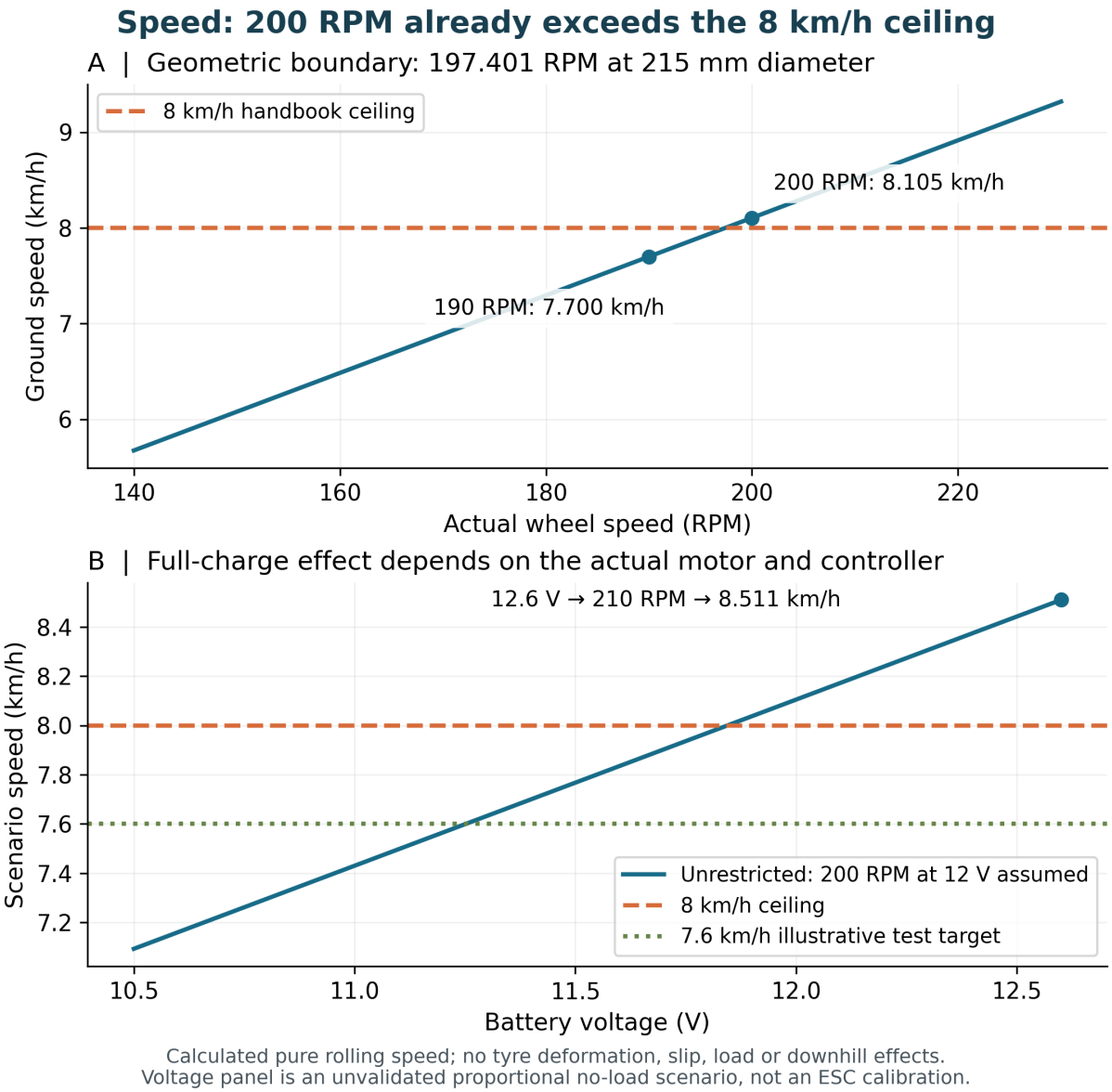


Figure 6: Calculated rolling speed versus wheel RPM and an explicitly conditional voltage-scaling scenario. These curves use the nominal 215 mm diameter; they are not measured speed traces.

Table 3: Calculated steady straight-climb lower bounds at assumed rolling resistance 0.10.

Slope	Required total force (N)	Wheel-output torque per wheel (N · m)	Minimum friction coefficient
0°	10.708	0.288	0.100
10°	29.139	0.783	0.276
15°	38.057	1.023	0.368
20°	46.685	1.255	0.464
25°	54.958	1.477	0.566
30°	62.813	1.688	0.677

At 20° and 2 km/h, wheel mechanical power is $Fv = 25.936$ W. If the total battery-to-wheel efficiency were 0.60, propulsion input would be 43.227 W. This efficiency is an illustrative assumption, not a measured motor/ESC map. Acceleration adds ma to longitudinal force; turning, sinkage, wheel unloading, impacts, and drivetrain losses can increase demand further. The tabulated torques are therefore lower bounds before design margin, not sufficient procurement specifications by themselves.

The continuous torque–speed–current curve is needed to determine whether each motor can sustain the required output at the desired speed. Stall torque is not an acceptable substitute for continuous torque. Similarly, a high advertised ESC peak rating does not establish reliable low-speed operation

under repeated skid turns. The mass-reduced comparison in the figure shows the direct effect of mass on the simplified force model, while preserving the uncertainty in terrain parameters.

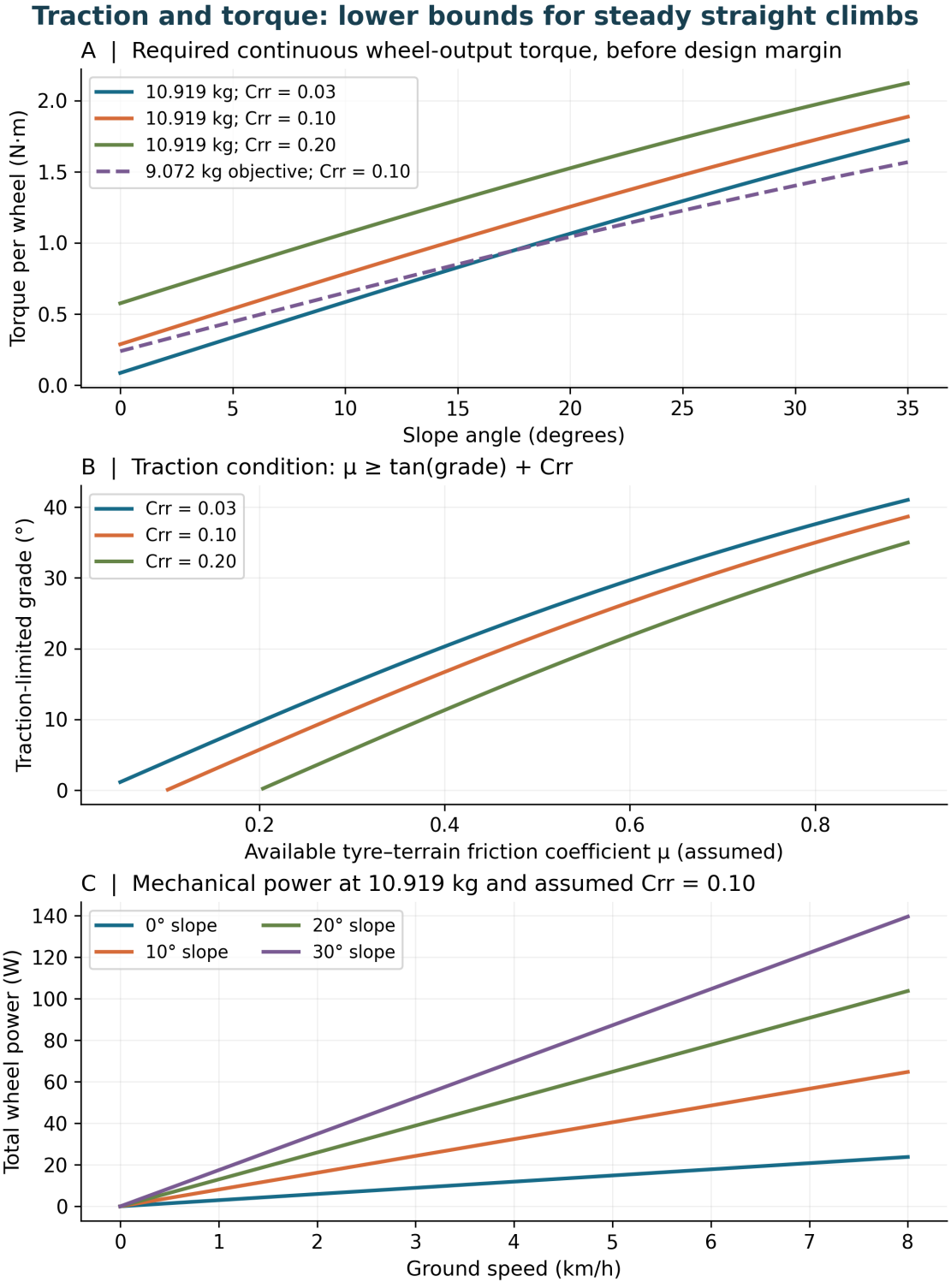


Figure 7: Analytical straight-climb sensitivities: wheel-output torque, friction-limited grade, and mechanical power. Rolling resistance, equal torque sharing and steady motion are declared assumptions. Turning and transient losses are excluded.

4.3. Static stability and changing contact geometry

For a centered center of mass at height h above an assumed rectangular support plane, first-order lateral and longitudinal tip angles are

$$\theta_{\text{lat}} = \tan^{-1}\left(\frac{T}{2h}\right), \qquad \theta_{\text{long}} = \tan^{-1}\left(\frac{L}{2h}\right).$$

Using the conditional $T = 0.540$ m and $L = 0.440$ m with an illustrative $h = 0.200$ m yields 53.47° and 47.73°, respectively. These are geometric thresholds for that idealized configuration, not approved operating slopes. The actual center of mass is unknown; suspension deflection, terrain contact changes, lateral acceleration, tire deformation, and payload movement can reduce available margin. If one wheel loses contact, the support polygon may become triangular and must be recomputed.

A useful first measurement is the loaded corner weight on a level surface. Four equal shares would each be 26.770 N, but the actual shares should be recorded with the installed batteries and task hardware. Controlled tilt measurements or a validated mass-property model can then locate the center of mass. The curve family below is more honest than reporting one "maximum slope" from an assumed height.

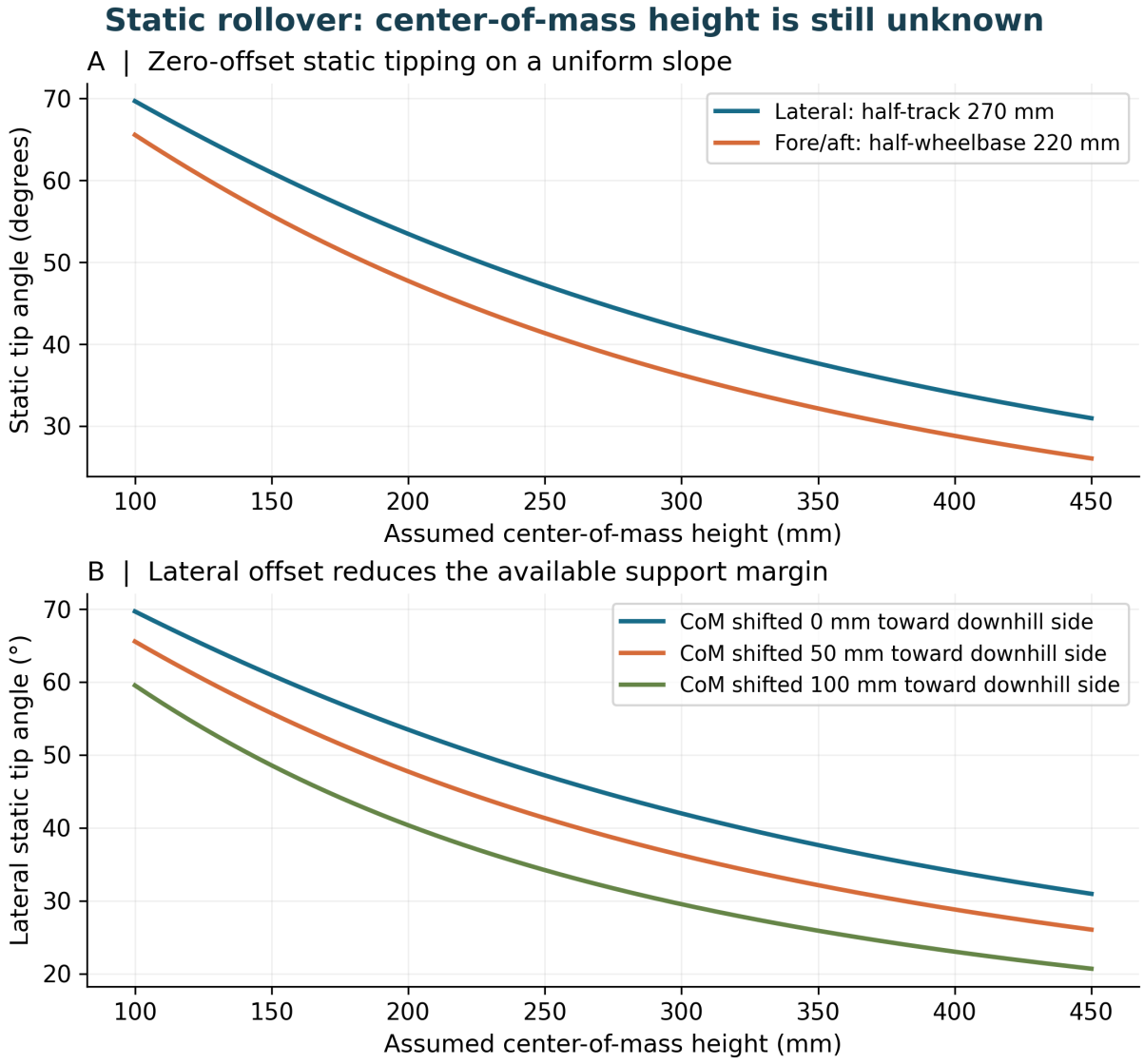


Figure 8: Conditional static tip-angle sensitivity to center-of-mass height. The support rectangle is inferred from the supplied external dimensions; the center-of-mass height is swept rather than measured.

5. Independent Suspension and Structural Integration

5.1. Local wheel rate and damping

For small wheel displacement z near a chosen ride height, let $\rho_i = dx_i/dz$ be the local motion ratio between spring compression and wheel displacement. With two springs of axial stiffness k_1 and k_2, the local wheel rate is approximately

$$k_w = k_1\rho_1^2 + k_2\rho_2^2.$$

This expression assumes locally constant motion ratios. If linkage geometry changes significantly or spring preload creates a geometric stiffness contribution, a full derivative of force with respect to wheel displacement is required. The installation angle alone does not generally define the motion ratio; attachment lever arms also matter. Two springs in the drawing cannot simply be treated as twice the wheel stiffness of one upright spring.

With one damper of axial coefficient c_d and motion ratio ρ_d, the local wheel damping is $c_w = c_d\rho_d^2$. For a linearized sprung corner mass m_s,

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k_w}{m_s}}, \qquad \zeta = \frac{c_w}{2\sqrt{k_w m_s}}.$$

The analysis sweeps spring rates and sprung masses because these quantities are not established by the purchase list. A total quarter-vehicle mass is not automatically a sprung mass: wheels, carriers, and portions of the motor assembly move differently from the chassis. The figures are sensitivity tools for selecting and measuring components, not predicted ride traces.

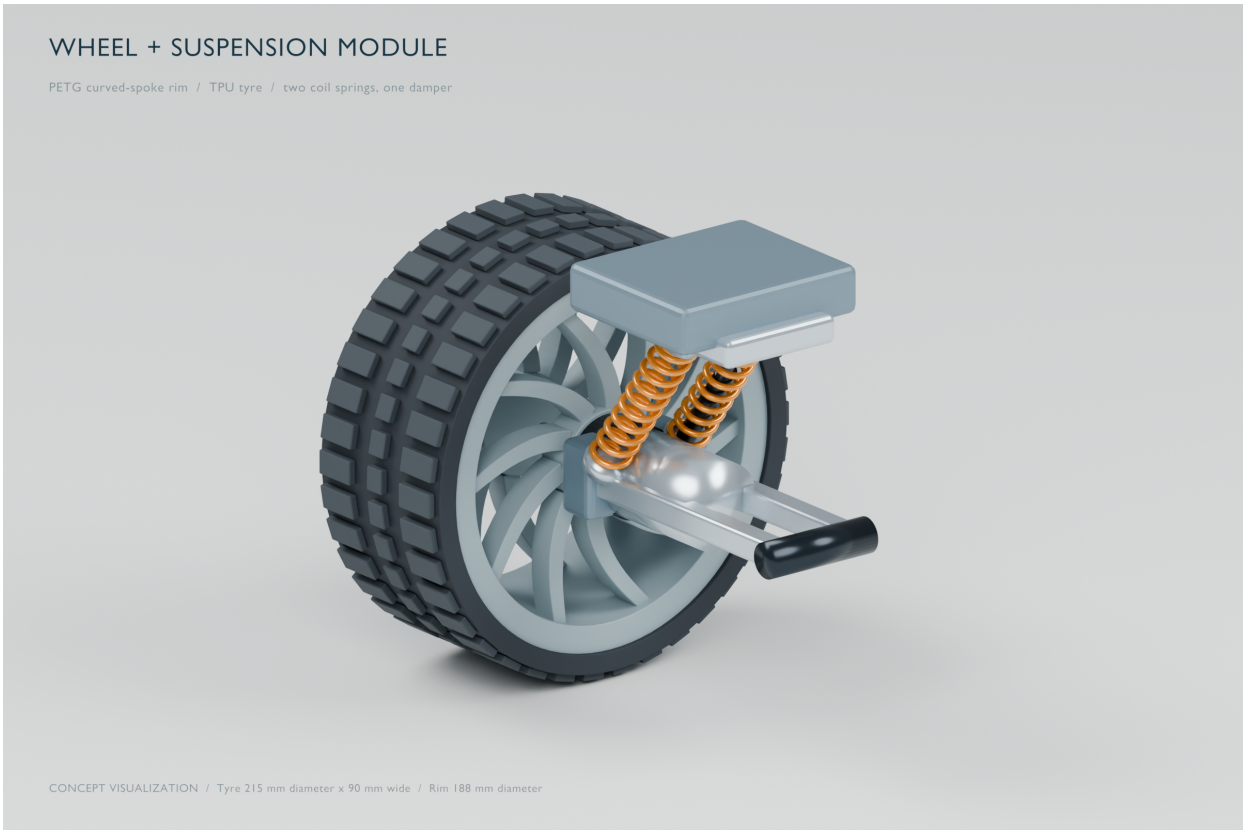


Figure 9: Blender illustration of one wheel module, showing the paired coils and single damper arrangement. The scene reconstructs the visible mechanism; spring rate, damping coefficient, bearing fits and full suspension travel remain to be measured or finalized.

5.2. Integration checks and practical consequences

The suspension should be tested first as a single loaded corner. Measure wheel displacement, spring compression, force, and return behavior over the intended travel. This reveals motion ratios, preload, friction, binding, and end-stop behavior without requiring a complete vehicle. A static test can estimate the wheel-rate curve, but damping needs a dynamic measurement. Recording free decay after a small displacement is useful only when the response is sufficiently linear and the sensors can resolve the motion.

At full assembly level, compare wheel loads and ground clearance at nominal ride height, one-wheel elevation, and diagonal articulation. The sensor mast moves with the chassis, so roll and pitch alter the camera and LiDAR viewing geometry. The task system should therefore perform its precise measurements from a settled position. Passive springs do not provide active leveling, and the present drawings do not establish an obstacle-height capability. Tire clearance must be checked at both compression and rebound, including wiring, fuse holders, and motor connectors on the moving carriers.

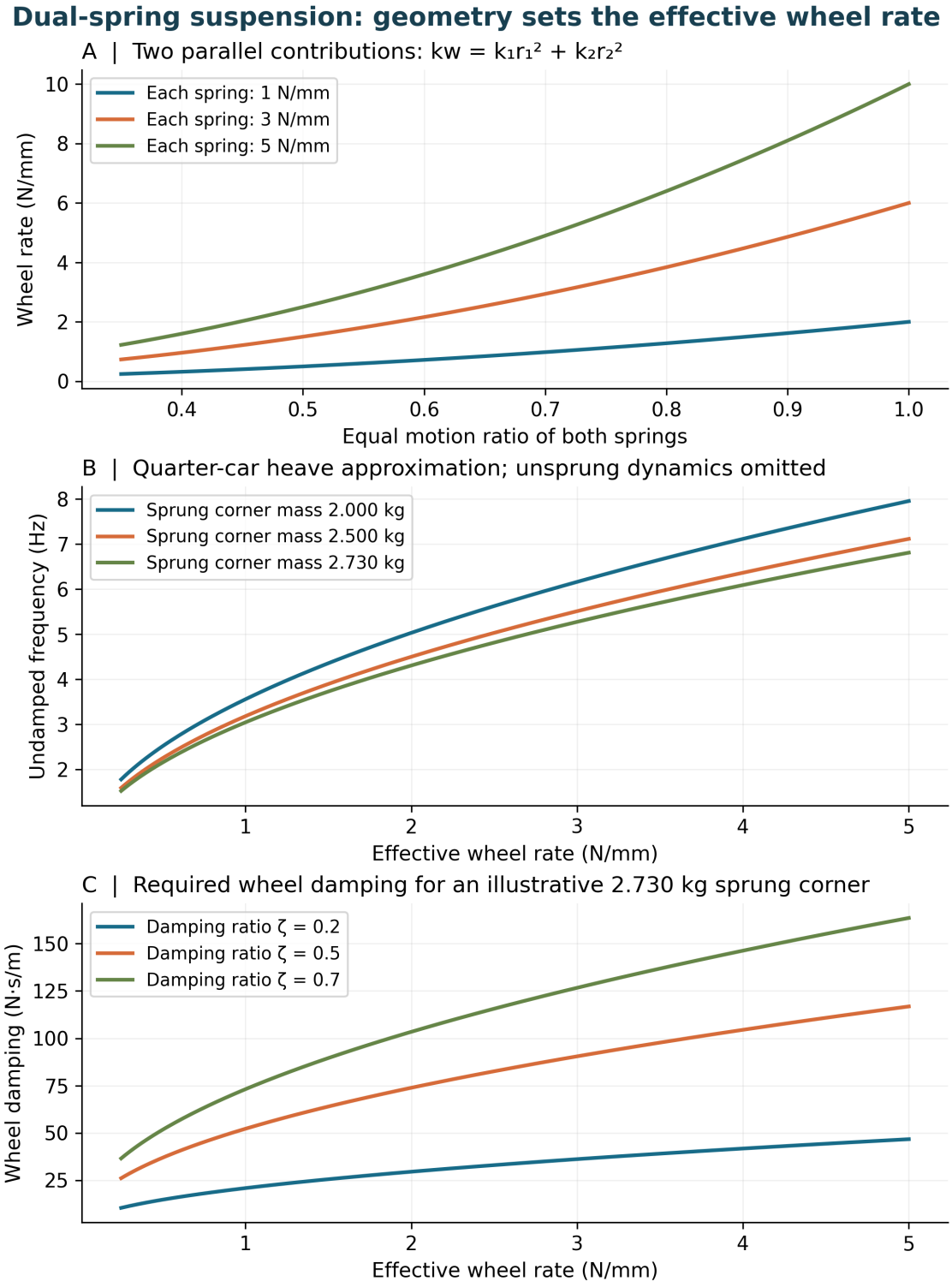


Figure 10: Local suspension sensitivity for two springs and one damper per wheel. Spring rates, motion ratios and sprung masses are illustrative parameter sweeps; none is a measured response of the proposed hardware.

6. Power Distribution, Protection, and Energy Budget

6.1. Separate propulsion and task supplies

The propulsion pack is a 3S, 5,000 mAh lithium-polymer battery; the task pack is a separate 3S, 2,200 mAh battery. Their nominal energies are

$$E_d = 11.1(5.0) = 55.50 \text{ Wh}, \qquad E_t = 11.1(2.2) = 24.42 \text{ Wh}.$$

The combined nominal energy is 79.92 Wh, but this total is not one interchangeable energy reservoir. Unused drive energy cannot sustain the task computer unless the architecture explicitly provides a suitable transfer path, which it currently does not. Each battery needs its own insulated enclosure, disconnect, restraint, and appropriately selected protection. Full-charge voltage is 12.6 V; nominal voltage is not the maximum regulator input condition.

The task pack supplies four separately regulated branches. The Pi branch uses the EP-0225 power arrangement to provide the specified USB-C power path for the Raspberry Pi 5 [27, 28]. A 5 V UBEC supplies the pan servo. Two DFR0379 regulators provide nominal 5.2 V rails for LiDAR and tilt-servo branches. The Raspberry Pi Camera Module 3 connects through the correct Pi 5 camera cable, while the display uses HDMI and USB as appropriate. The selected hardware revision and connector polarity must be checked before power is applied.

The PCA9685 breakout board has a logic supply and a separate external servo-power arrangement. Its regulated positive servo rail must not accidentally join the independently supplied pan and tilt rails. The Pi-side I^2C logic and pull-ups remain at 3.3 V. An SN74AHCT125 powered at 5 V can provide a TTL-compatible level interface for the pan control signal [29]. The tilt servo's acceptance of the intended logic level still requires verification; a supply-voltage rating alone does not establish its input-high threshold. All signal returns must follow the designed task return path.

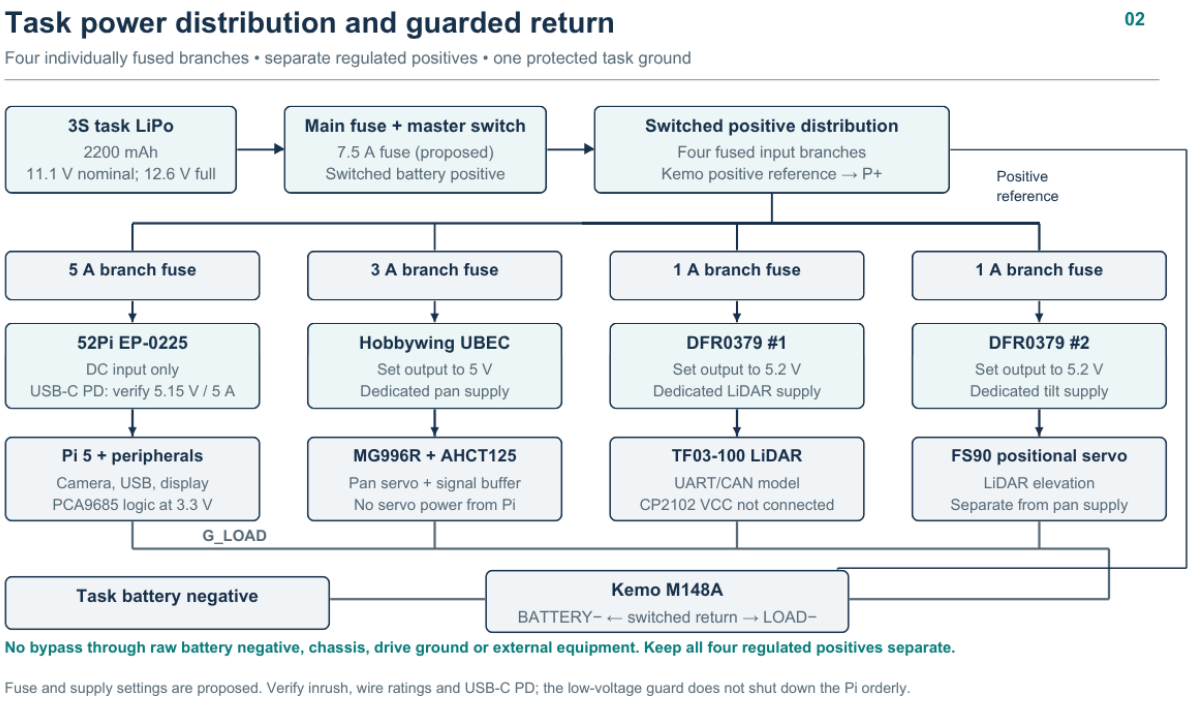


Figure 11: Editable Canva task-power diagram. The separate positive rails and the low-side battery guard are explicit. Branch labels describe proposed hardware, not measured current or validated fuse coordination.

6.2. Low-side guard and shutdown behavior

The M148A guard switches the negative return path [30]. Its BATTERY− terminal belongs to the raw task-battery negative, while all protected task loads return through LOAD−. An unintended USB, chassis, instrument, or drive-system ground connection can bypass that switching path. Bench

equipment must therefore be connected with awareness of the intended return topology. The positive reference follows the switched positive distribution as specified by the final wiring design.

A pack-level guard is not a cell-balancing system and does not replace per-cell monitoring. Nor does sudden power removal constitute orderly operating-system shutdown. The Pi must finish writing photographs and event records before its supply is removed. A practical operating procedure needs a verified voltage threshold, sufficient reserve for shutdown, and testing under transient servo and USB loads. Because a pack threshold does not directly reveal an individual weak cell, the design should include the planned per-cell alarm and pre-run battery checks.

Electrical integration tests should record the Pi input voltage, regulator outputs, task-pack current, and relevant return-path voltages while the pan servo starts, the camera captures, and the display is active. A successful idle boot does not demonstrate immunity to servo transients or radio-frequency interference. Active cooling should be installed and its wiring secured; CPU throttling and converter heating must be checked during the complete mission sequence.

6.3. Illustrative current and runtime envelopes

For an output branch power P_o, battery voltage V_b, and assumed converter efficiency η, input current is

$$I_b = \frac{P_o}{\eta V_b}.$$

Input current must be compared with an input fuse; output current must be compared with the output rail and load wiring. Mixing those quantities produces misleading protection calculations. The following simultaneous-load example uses $V_b = 10.5$ V and $\eta = 0.85$. It is a planning scenario, not an instrumented load profile.

Table 4: Illustrative task-branch load and battery-side current budget.

Task branch	Illustrative output power (W)	Calculated battery-side current (A)	Proposed input fuse (A)
Pi and USB branch	25.75	2.885	5
Pan-servo branch	15.00	1.681	3
LiDAR branch	2.00	0.224	1
Tilt-servo branch	5.20	0.583	1
Total	47.95	5.373	7.5 main

The nominal difference between 7.5 A and 5.373 A is not a validated margin against nuisance operation or cable damage. Fuse time–current curves, wire ampacity, ambient temperature, regulator startup current, servo stall current, and source fault current remain relevant. Regulators can also limit or oscillate before a fuse operates. The protection scheme must be assessed as a coordinated circuit.

Assuming 80% usable nominal pack energy over a 12-minute mission gives average battery-power budgets of 222 W for propulsion and 97.68 W for tasks. With 85% task conversion efficiency, the latter corresponds to 83.03 W of average output power. These are arithmetic energy ceilings under assumed usable fractions. They do not demonstrate that either pack can provide a particular current without sag, that the motors can tolerate the heat, or that a regulator can deliver that output continuously.

For a constant 100 W propulsion draw, the same model gives 26.64 minutes. A constant 25 W task output at 85% efficiency gives 39.85 minutes. Real mission endurance requires a measured duty cycle, temperature, cell balance, converter efficiency, and shutdown reserve. The graphs retain these distinctions instead of treating an optimistic capacity calculation as a completed runtime test.

7. Packet-Commanded Imaging and LiDAR Ranging

7.1. Task sequence and receiving architecture

The supplied handbook describes 4 ft × 4 ft white target boards carrying a filled red circle, blue square, or green triangle. The red circle diameter and blue square side are 12 inches; the equilateral green triangle has a 12-inch height. Targets are placed within 50 ft of the specified starting point. The task

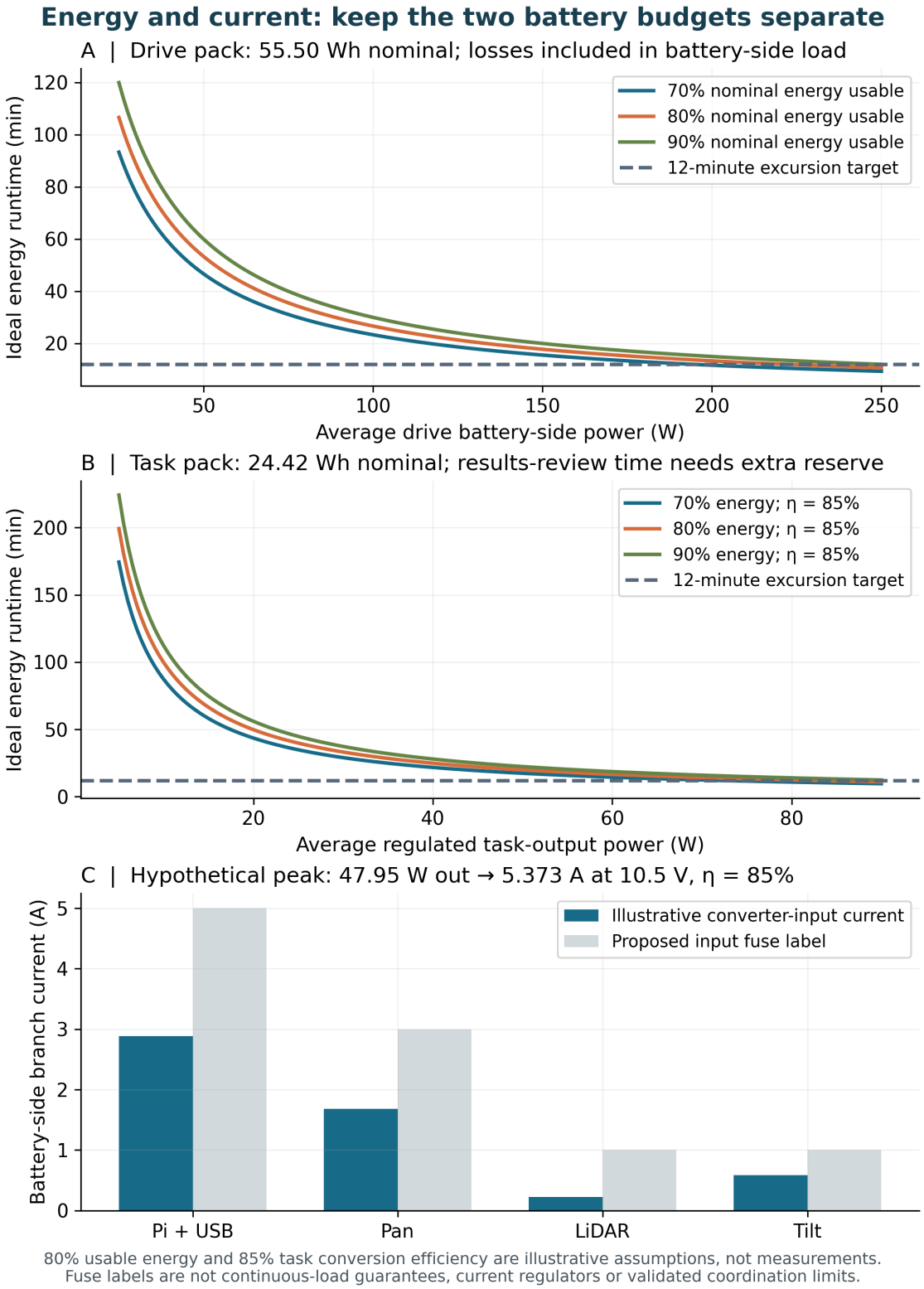


Figure 12: Calculated energy and current scenarios for separate drive and task packs. Assumed usable energy and conversion efficiency are shown; the figures do not represent measured battery discharge or load traces.

commands select pan motion, photography, color filtering, and distance measurement [1]. The software must therefore preserve command order and state even when successive commands repeat a letter.

The receiving chain uses a VHF antenna, an RTL-SDR V3 receiver, demodulation, and Dire Wolf AX.25/APRS decoding [31]. The operating description specifies 144–148 MHz and 1200 baud; the final receiving frequency and packet addressing must follow NASA's released configuration [1]. The program should accept validated decoded command tokens through an adapter, rather than hard-code a guessed APRS payload format. A successful software decode in a laboratory does not establish outdoor reception reliability in the presence of motors, switching converters, USB devices, and nearby transmitters.

Table 5: Received task-command semantics and execution state.

Token	Required operation represented in the executor	State consequence
AAA	Begin a task sequence	Open a new record after successful initialization
B / C	Turn field of view 30° right / left	Update pan only after successful motion
D	Capture and save a new photograph	Use current filter; attach capture date/time
E / F	Select red / blue filter	Replace the prior filter selection
G	Select no filter	Restore normal-color output
H	Measure target distance	Record a fresh, valid result in feet
ZZZ	End the sequence	Finalize records for onboard review

The command letters and 30-second interval come from the task specification; the prototype's internal 25-second service budget is an engineering proposal, not a measured task latency. The reference timeout applies after dispatch begins. Queue waiting and cancellation acknowledgement are additional, so 25 seconds is not a hard packet-arrival-to-result deadline. Reception should continue while a command executes. A queue separates the receiver from motion, capture, and ranging so that slow operations do not silently discard later commands. Repeated B commands may legitimately request successive 30° increments and must not be discarded merely because their payload letters match.

7.2. Fresh photographs and filter semantics

The camera is mounted on a bearing-supported pan assembly. The commanded field-of-view change should correspond to measured physical camera rotation. A hobby-servo pulse command alone does not establish angular accuracy under backlash, cable load, or asymmetric mechanical stops. The total pan range required by an unknown command sequence is not established in the supplied description; consequently, a nominal 180° servo must not be declared sufficient without resolving the permitted sequence and calibrating usable travel.

For each D command, the program requests a new camera frame after the selected orientation has settled. The frame's capture timestamp is obtained from a correctly initialized clock, and the resulting image is stored with the required date and time. The RTC cell preserves time across power loss but does not itself guarantee that time was initially set correctly. Unique file identifiers prevent a repeated photograph command from overwriting earlier evidence.

Software red or blue filtering should be defined and demonstrated consistently. One candidate channel-isolation transform is

$$I_{\text{red}} = (R, 0, 0), \quad I_{\text{blue}} = (0, 0, B), \quad I_{\text{normal}} = (R, G, B).$$

This describes the proposed software interpretation; it is not an assertion that the handbook mandates this exact pixel operation or that software channel isolation is identical to a physical optical filter. The chosen implementation should be confirmed against the task's intended filter interpretation. A new filter replaces the previous one, and "no filter" means ordinary color, not grayscale. The original unfiltered frame is the appropriate input for the optional classifier, because applying a red or blue output filter would destroy color evidence needed for three-class target recognition.

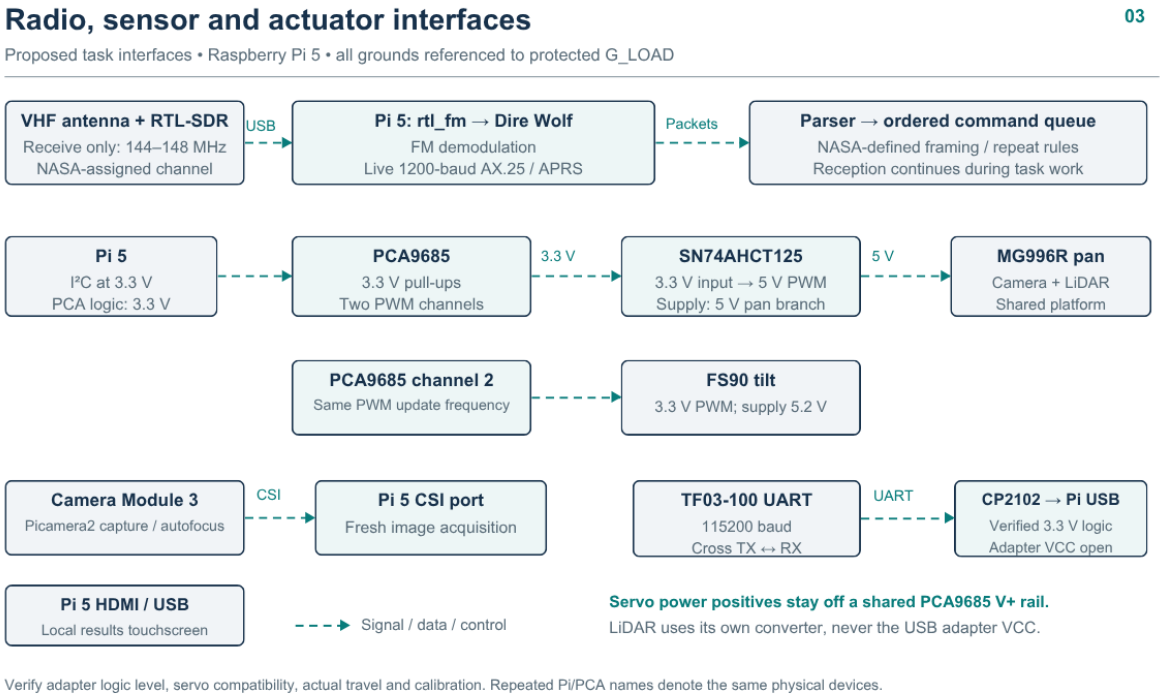


Figure 13: Editable Canva task-interface diagram showing the receiving chain, computer, camera, LiDAR, servo control, storage and display. This is an integration specification; exact device revision and pin-level checks remain necessary.

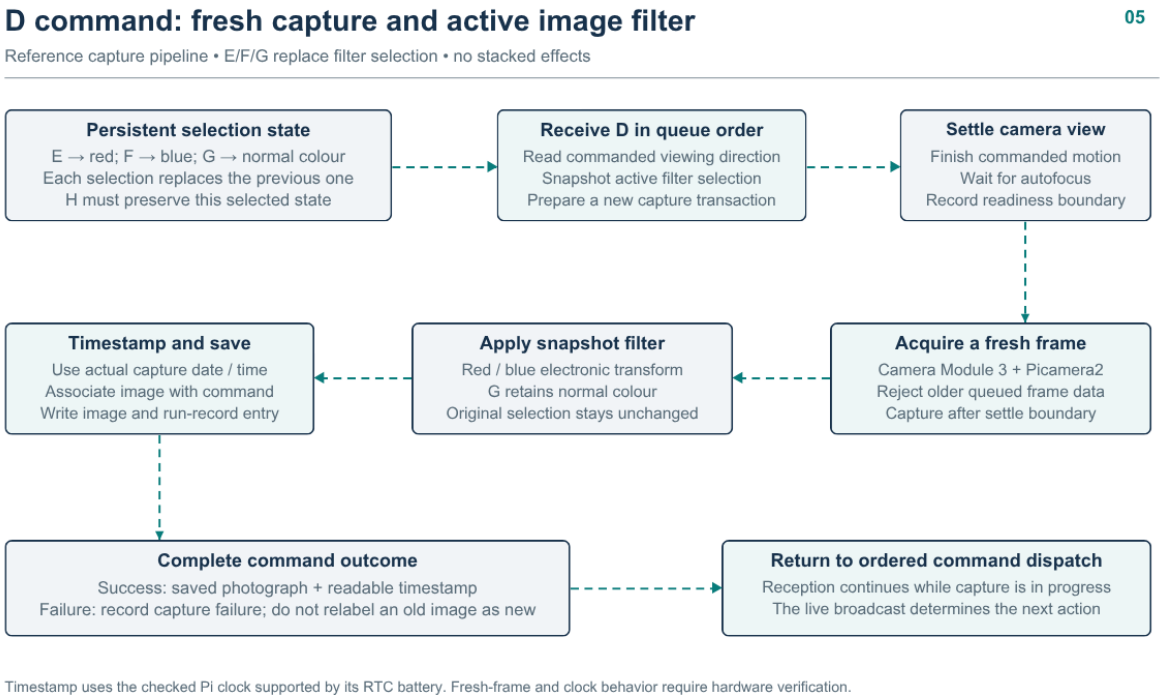


Figure 14: Editable Canva capture pipeline. The command queue, current filter, settled view, fresh frame, timestamp, storage and onboard review are represented separately so that an image can be traced to its command.

7.3. Pixel sampling and camera calibration

The Camera Module 3 Standard product brief specifies a 4608 × 2592 sensor image and a nominal horizontal field of view of 66° [32]. A full-frame pinhole approximation gives focal length in pixels

$$f_x = \frac{N_x}{2\tan(\alpha_x/2)}, \qquad p = \frac{f_x w}{Z},$$

where $N_x = 4608$, $\alpha_x = 66°$, w is feature width, and Z is optical-axis depth to a fronto-parallel feature, rather than general Euclidean sensor-to-target range. For a 12-inch width at 50 ft, $p \approx 70.96$ pixels. Scaling the same complete view to 1920 pixels across gives approximately 29.57 pixels. This is a sampling estimate rather than a recognition guarantee. Lens distortion, focus, exposure, motion blur, compression, frame cropping, and the actual board pose affect the image presented to an algorithm.

Table 6: Ideal sampling of a 12-inch feature using the complete camera view.

Range	Angular width of a 12-inch feature	Full 4608-pixel width estimate	Same full view scaled to 1920 pixels
10 ft	5.725°	354.78 px	147.83 px
25 ft	2.292°	141.91 px	59.13 px
50 ft	1.146°	70.96 px	29.57 px

The triangle's specified height is 12 inches; its equilateral side is $2h/\sqrt{3}$, not 12 inches. The table uses a generic 12-inch feature dimension and should not be interpreted as assigning identical bounding boxes to all shapes. Camera calibration should be performed at the chosen capture mode and focus configuration with target views spanning the usable image area [18]. Reprojection residuals quantify the fitted camera model; they do not independently establish LiDAR alignment or task accuracy.

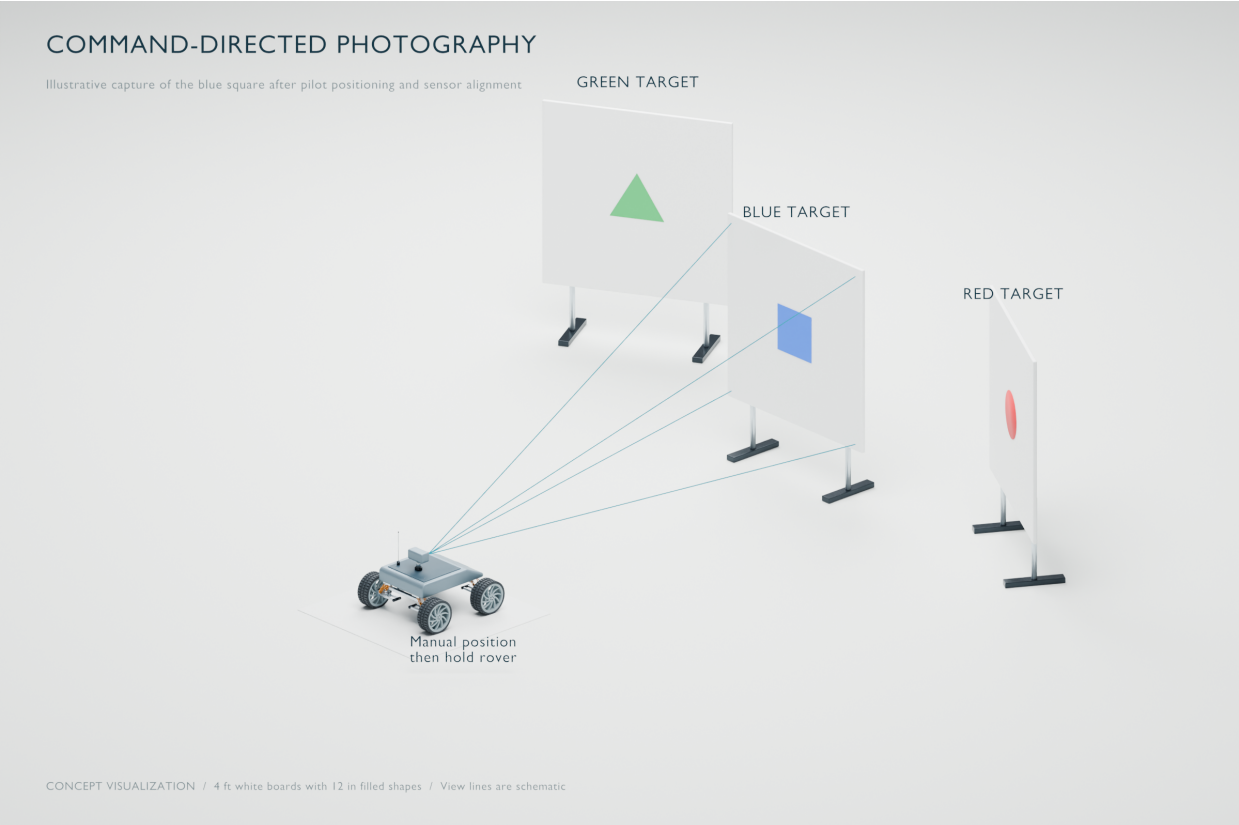


Figure 15: Blender illustration of the photography task with the rover stationary and illustrative target placements. The scene is a visualization of the intended operation, not a rendered dataset used for the classification experiment and not an observed competition run.

7.4. Single-beam ranging and reference geometry

The TF03-100 is a single-beam range sensor. Its listed field of view is 0.5° and its performance is conditional on target reflectivity and environmental conditions [33]. Treating 0.5° as a full cone angle for a geometric illustration gives

$$b(R) = 2R\tan(0.25^\circ).$$

Here R denotes the nominal perpendicular distance to the target plane; the footprint calculation assumes the beam is normal to that plane. For a pointing error δ, the ray's slant distance to the same plane is $R\sec\delta$. At 50 ft, this construction produces a nominal width of about 133.0 mm. It is not a verified beam-profile measurement. A pointing error δ displaces the ray laterally by $R\tan\delta$. At 50 ft, 1° produces 266.0 mm and 3° produces 798.7 mm. The latter exceeds the half-width of a centered 4 ft board. Thus a stable numerical range can still be associated with the wrong surface.

By contrast, if the ray still intersects the same flat plane, the slant-minus-normal difference is $R(\sec\delta - 1)$, only 20.9 mm at 3° and 50 ft. This comparison identifies board association and boresight alignment as primary integration problems. Repeatedly averaging a wrong-surface return does not fix them.

The ranging adapter should aim at the intended board, verify serial framing and checksum, reject invalid or stale values, collect a bounded set of fresh samples, and report failure explicitly if no valid board-associated measurement is obtained. A CP2102 serial adapter must use a compatible logic interface, and the LiDAR receives its specified regulated supply separately. A physical calibration procedure must determine camera intrinsics, the camera-to-LiDAR transform, servo-angle mapping, mounting offsets, and the effect of sensor tilt. Multi-beam calibration papers provide useful principles [19, 20], but their observation models cannot simply be transplanted to a single beam.

Let $\mathbf{s}$ be the sensor position relative to a chosen rover reference, pending confirmation of the reporting datum, and $\mathbf{u}$ the calibrated unit beam direction. A measured range ρ gives target position

$$\mathbf{q} = \mathbf{s} + \rho\mathbf{u}.$$

If the required reported quantity is Euclidean reference-to-target distance, it is $\|\mathbf{q}\|$; if the requirement is horizontal distance, the horizontal projection is used. A fixed scalar subtraction is generally wrong while the head pans. The correct reference definition must be confirmed, and the human-powered division's ±3 ft allowance must not be assigned to the RC task. Manufacturer accuracy figures are likewise not NASA judging tolerances. At 50 ft, the TF03-100 datasheet's 1% accuracy specification corresponds to ±0.5 ft under its stated 90%-reflective-white-board conditions; this does not characterize the complete mounted system.

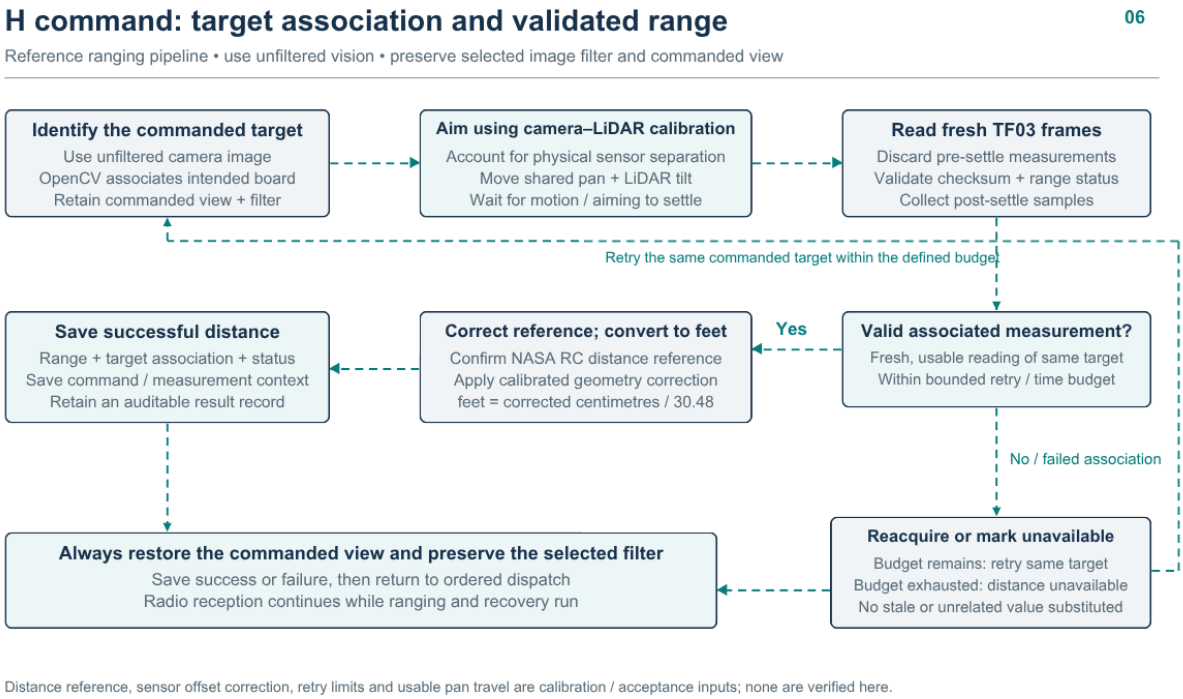


Figure 16: Editable Canva ranging pipeline with target association, calibration, frame validation, fresh sampling, bounded retry and explicit unavailable-result handling. A plausible distance alone is insufficient evidence of a correct target measurement.

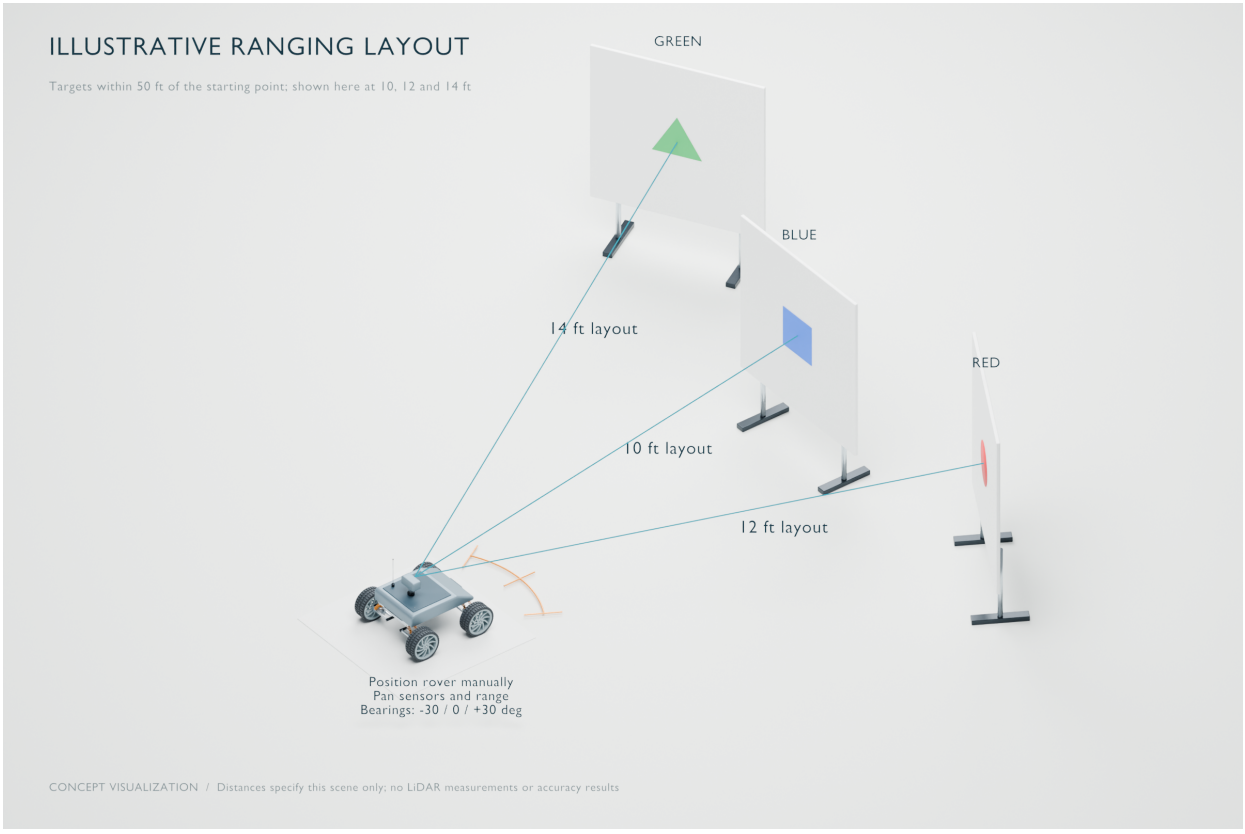


Figure 17: Blender ranging illustration. The beam and annotated board layout are schematic visualization elements; distances in the scene are illustrative, not recorded TF03 measurements or a surveyed NASA course.

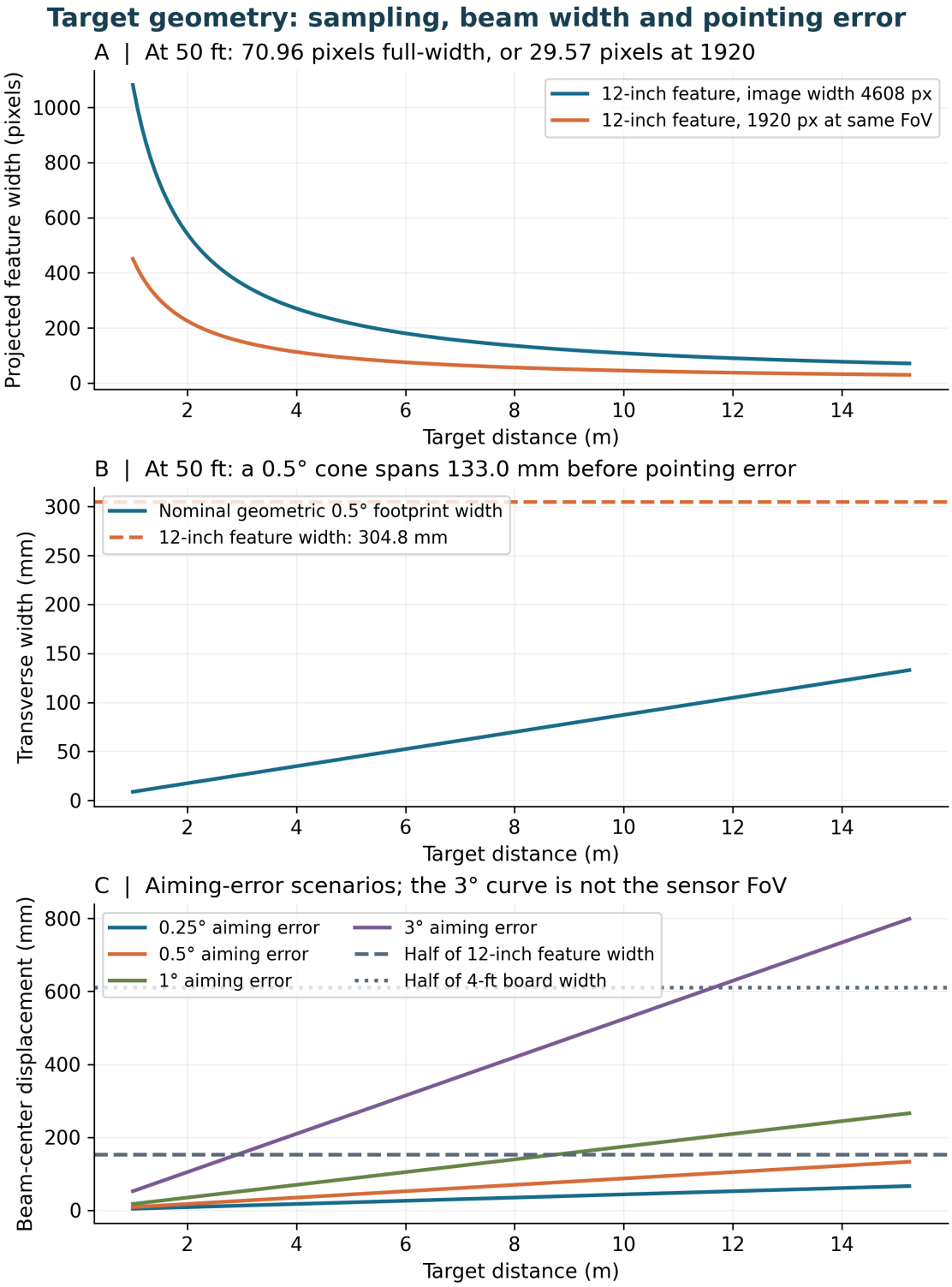


Figure 18: Calculated camera sampling, nominal beam-width geometry and aiming-error sensitivity. These are deterministic geometric calculations with declared sensor specifications, not calibration residuals from the rover.

8. Reference Software and Command Verification

8.1. State model and implementation boundary

The accompanying Python reference executor uses WAITING, ACTIVE, and FINALIZED states. A successful AAA record-opening operation establishes ACTIVE. Filters are state variables; motion updates the commanded pan only after its adapter succeeds. A serialized acceptance path prevents overlapping commands from racing. A hardware error or timeout records an explicit result and inhibits dependent motion, photography, and ranging until the system has been deliberately recovered. Record finalization remains possible so that partial evidence is retained. The reference exposes a recovery-required flag but does not implement physical recovery or an automatic flag-clearing procedure; these belong to the hardware integration.

The executor receives already validated tokens. Radio packet addressing, exact payload syntax, retransmission identity, and source and address validation supported by the released protocol belong to the receiving adapter and must follow NASA's released specification. The code does not fabricate that wire format. Its event list is an in-memory trace, so the deployed adapter must additionally persist the required records. Cancellation of an asynchronous call cannot by itself guarantee that a physical servo stops; the hardware interface must define bounded operations, recovery, and safe behavior after communication loss.

The illustrative ±90° pan bounds in the reference code are test parameters. They are not a conclusion that 180° travel satisfies all possible commands. Out-of-range turns are rejected and logged, rather than silently clamped or wrapped. This behavior exposes a design deficiency instead of disguising it as successful execution. A complete competition implementation must resolve mechanical range before deployment.

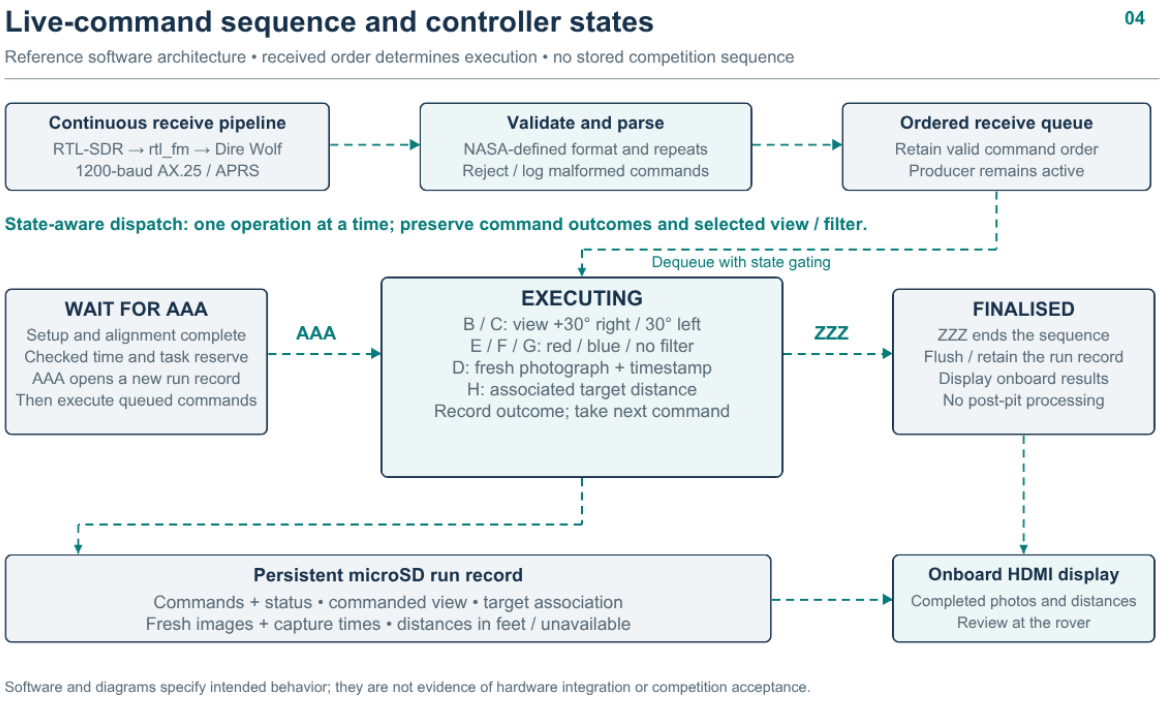


Figure 19: Editable Canva command-state and fault-flow diagram. Reception and execution are decoupled, and invalid, out-of-range or failed operations remain visible in the record.

The following extract illustrates state replacement, bounded motion, and explicit unit conversion. It is explanatory code from the hardware-independent design, not a complete radio or motor driver.

```
if token in {"E", "F", "G"}:
    image_filter = {"E": "red", "F": "blue", "G": "normal"}[token]
elif token in {"B", "C"}:
    target = pan_deg + (30.0 if token == "B" else -30.0)
    if not pan_min <= target <= pan_max:
        return {"status": "rejected", "reason": "pan_range"}
    await hardware.pan_to(target)
    pan_deg = target
elif token == "H":
    range_cm = await hardware.measure_target(pan_deg)
    valid = (isinstance(range_cm, (int, float))
             and not isinstance(range_cm, bool)
             and math.isfinite(range_cm) and range_cm > 0)
    if not valid:
        return {"status": "distance_unavailable"}
    return {"status": "range_saved", "distance_ft": range_cm / 30.48}
```

The conversion factor is exact because 1 ft = 30.48 cm. A range adapter must return a correctly referenced value in the declared unit; the executor cannot repair a wrong datum or mixed millimeter/centimeter input merely by dividing. Successful execution should also preserve the requested camera pan and filter across any temporary LiDAR targeting motion. The current software checks logical state preservation; physical restoration needs hardware measurement.

8.2. Executed checks and deployment tests

The verification script executes 30 focused checks with mocked hardware. Cases include commands outside an active sequence, repeated 30° turns, filter replacement, distinct photo requests, valid and invalid ranges, unknown tokens, sequence finalization, queue order, out-of-range motion, timeout inhibition, record-opening failure, concurrent acceptance, and cancellation. Boolean values are rejected as distances even though Python treats booleans as integer subclasses. A photo-adapter result cannot silently overwrite the executor's own status.

These checks are software unit-level evidence. They do not exercise an RF receiver, physical servo, camera exposure, actual storage device, or LiDAR target. They nevertheless provide a reproducible foundation for integration because they define failure behavior before physical tests. The next stage is to replace mocks with device adapters while preserving the same event schema and to replay complete sequences under controlled fault injection.

A useful commissioning sequence starts with the rover mechanically supported, then proceeds to stationary task operation, and only later to combined drive and task tests. Inject malformed serial frames, remove the LiDAR briefly, force an image-write error, and test recovery after a supply interruption. Confirm that the onboard display exposes missing data rather than substituting old images or zero-valued ranges. The final acceptance demonstration must show fresh results after an unknown sequence, with no external editing or return to the pit.

9. Synthetic Evaluation of Learning-Assisted Target Recognition

9.1. Dataset and experimental separation

The optional recognition extension was evaluated on an executed synthetic benchmark because no physical camera recordings or target trials were supplied. The task is four-way classification of a supplied board crop: red circle, blue square, green triangle, or no target. It does not test full-frame board detection, autonomous steering, or physical Raspberry Pi integration. A procedural image generator creates 64×64 RGB crops containing a projectively transformed light board, generated background, and a marker or negative example. It renders at double resolution before downsampling, blur, and additive noise.

There are 2,400 images in 600 scene groups. Each group contains all four labels with shared nuisance parameters, and each entire group belongs to one split. This prevents the same pose, background, and lighting configuration from leaking across training and evaluation. The negative class contains blank

boards or one of six incorrect color–shape combinations. Those are deliberately difficult negatives, not additional NASA target classes. Positive shapes have matched sampled area before projection, reducing an avoidable area cue; the generated geometry is not a metrically exact reproduction of all handbook board dimensions.

Table 7: Scene-disjoint synthetic dataset allocation.

Split	Scene groups	Images	Purpose
Training	300	1,200	Fit feature standardization and SVM parameters
Validation	100	400	Select declared hyperparameters
Nominal test	100	400	Evaluate unseen scenes from nominal generator ranges
Shifted test	100	400	Evaluate altered backgrounds and stronger corruptions

Each test set has 100 examples per label. The generator varies board size, rotation, perspective, marker location, color gain, exposure, illumination gradients, blur, and noise. The shifted set uses held-out checkerboard and diagonal-stripe background families, broader rotations, stronger color changes, and increased blur and noise. All split seed ranges are disjoint and all final image hashes are unique. This design follows the motivation of appearance randomization [21, 22], but the altered test generator is not a real-world domain.

Table 8: Selected appearance ranges in the nominal and shifted generators.

Nuisance parameter	Nominal range	Shifted range
Board width / image width	0.72–0.88	0.64–0.83
Rotation	$-30°$ to $+30°$	$-55°$ to $+55°$
Per-corner projective jitter	± 0.045 image widths	± 0.10 image widths
Exposure multiplier	0.72–1.20	0.38–1.40
Per-channel gain	0.85–1.15	0.58–1.40
Blur standard deviation	0–0.65 output pixels	0.65–1.65 output pixels
Noise standard deviation	0–4 gray levels	3–12 gray levels

Because several nuisance factors change together, the experiment cannot attribute performance loss to one corruption. The source package includes the exact sampled parameters and RGB hashes for every example, allowing the evaluation to be reproduced or separated into later controlled ablations.

9.2. Features, classifiers, and model selection

Three compact methods are compared. The HSV-and-geometry baseline thresholds colored regions, performs morphological cleanup, identifies connected components, and evaluates convex-hull circularity, oriented-box fill, and a heuristic solidity measure. Validation selects three geometric cutoffs from 27 declared combinations. This method is transparent and uses task-specific shape knowledge, but its thresholds can be affected by lighting, occlusion, and low pixel resolution.

The first learned method uses a normalized 18×3 hue–saturation histogram and eight value bins, giving 62 features. The second appends 1,764 grayscale HOG features, giving 1,826 total inputs. The HOG variant uses nine unsigned orientation bins, 8×8-pixel cells, overlapping 2×2-cell blocks, and L2-Hys normalization. Orientation votes are interpolated; spatial vote interpolation is omitted. It is therefore a documented HOG variant rather than an exact reimplementation of Dalal and Triggs [34].

Both learned methods use linear support-vector classifiers [35]. For each one-versus-rest binary problem, the squared-hinge formulation has the form

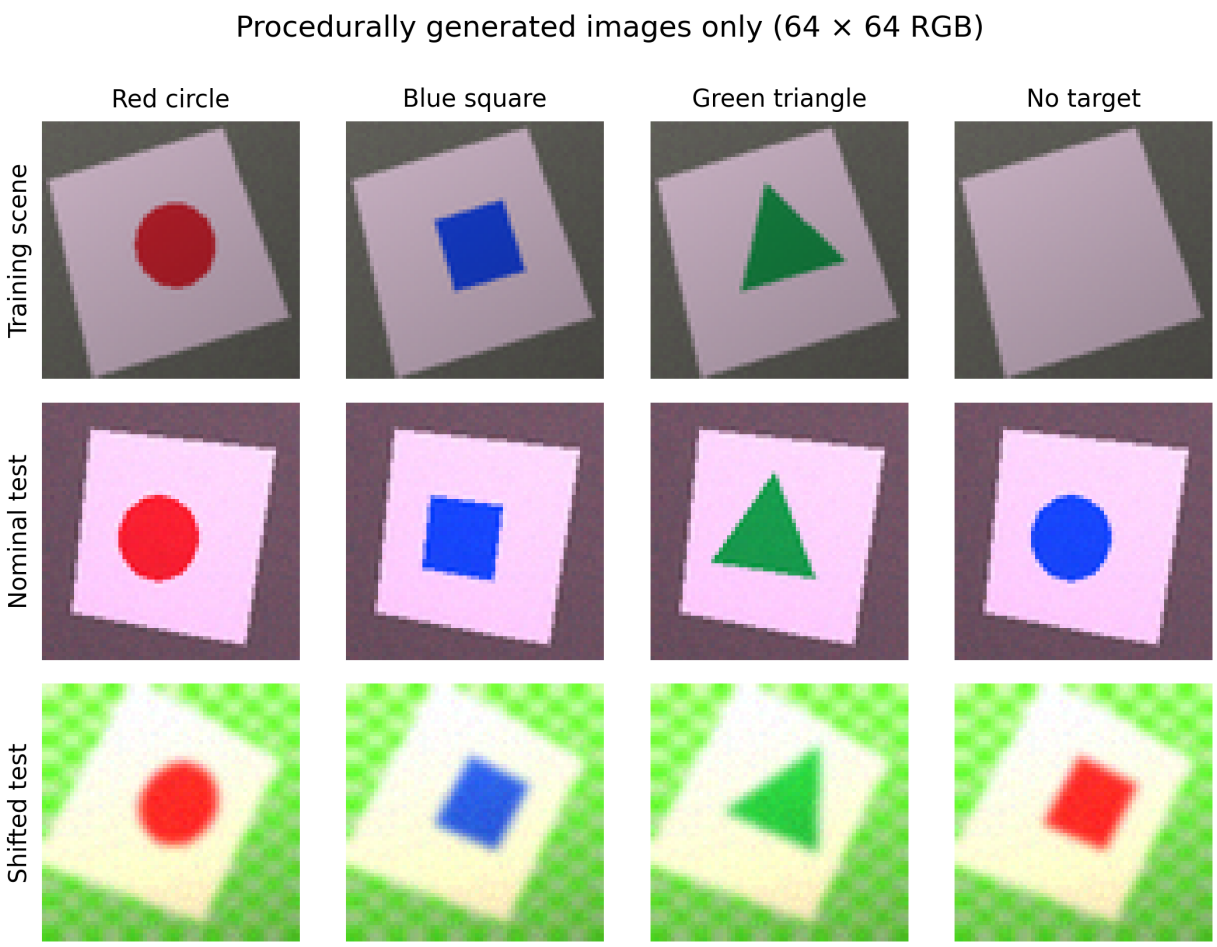


Figure 20: Examples from the executed synthetic dataset. Nominal and shifted board crops include target markers and no-target negatives. These are procedurally generated images, not photographs from the rover camera or Blender task scenes.

$$\min_{\mathbf{w},b} \frac{1}{2}\left(\|\mathbf{w}\|^2 + b^2\right) + C\sum_i \max\left(0, 1 - y_i(\mathbf{w}^\mathsf{T}\tilde{\mathbf{x}}_i + b)\right)^2,$$

where $\tilde{\mathbf{x}}_i$ is standardized using training-set statistics and $y_i \in \{-1, +1\}$. With `fit_intercept=True` and the default `intercept_scaling=1`, `LinearSVC` penalizes the intercept through a constant synthetic feature. Multiclass prediction selects the largest class decision value. Decision values and gaps between them are not calibrated probabilities.

For each SVM, validation macro-F1 selects $C \in \{0.0001, 0.001, 0.01\}$, with ties choosing the smallest value. The final bounded grid follows a computational pilot in which larger-C fits were slow; no HOG test score informed that change. Interrupted pilots emitted color-only evaluation metrics; the grid change was based on solver runtime. The experiment is exploratory rather than preregistered. Models are fitted on training examples only and are not refitted on validation data. The declared procedure can be expressed compactly as follows.

```
best = None
for C in (0.0001, 0.001, 0.01):
    model = make_pipeline(
        StandardScaler(),
        LinearSVC(C=C, dual=False, max_iter=2000,
                  random_state=1703))
    model.fit(X_train, y_train)
    score = f1_score(y_valid, model.predict(X_valid), average="macro")
    if best is None or score > best[0]:
        best = (score, C, model)  # ascending C resolves ties
selected_model = best[2]
nominal_predictions = selected_model.predict(X_test)
shifted_predictions = selected_model.predict(X_shift)
```

The displayed listing summarizes model selection; the complete generator, HOG implementation, geometric classifier, metrics, and reproducibility controls are in the accompanying source. For each class k,

$$F_{1,k} = \frac{2TP_k}{2TP_k + FP_k + FN_k}, \qquad F_{1,\text{macro}} = \frac{1}{4}\sum_{k=1}^{4} F_{1,k}.$$

Accuracy and macro-F1 are recomputed from saved predictions. The 95% percentile intervals use 1,000 bootstrap resamples of complete four-image scene groups. These intervals represent scene-sampling variation conditional on the fitted models; they do not include training randomness, alternate generator choices, or physical-camera uncertainty.

9.3. Results and the value of a negative comparison

Table 9: Executed classification results on the two synthetic test sets.

Method	Nominal accuracy	Nominal macro-F1	Shifted accuracy	Shifted macro-F1
HSV + geometry	95.00%	0.9512	71.75%	0.7343
HSV histogram + SVM	82.00%	0.7892	61.75%	0.5838
HOG + HSV + SVM	82.00%	0.7890	48.50%	0.4784

The explicit geometric baseline performs best on both test sets. Adding HOG provides no nominal-accuracy improvement over the color histogram and reduces shifted accuracy by 13.25 percentage points. Both learned models correctly reject only 34 of 100 nominal no-target examples. A report restricted to positive-marker accuracy would therefore conceal a significant failure mode. The selected C values are 0.001 for the color model and 0.0001 for the combined model.

The nominal accuracy interval is 91.00–98.25% for the geometric baseline and 79.50–84.75% for the HOG-plus-color model. Under the shifted generator, the corresponding intervals are 65.50–78.00% and 43.49–52.75%. These are group-bootstrap intervals, not field success bounds. The result does not establish that geometry will always outperform learning; it shows that adding this feature representation and linear classifier did not improve this declared benchmark.

The HOG-plus-color model artifact occupies 98,188 bytes. Feature extraction and one-image prediction take a median of 3.466 ms and a 95th percentile of 10.542 ms across 300 timed calls on a shared x86-64 Xeon host, after warm-up and with a one-thread BLAS limit. The timing excludes camera acquisition, image decoding, board localization, and control. It is not Raspberry Pi 5 timing. A compact artifact and short host inference time do not compensate for poor negative rejection.

9.4. Operational boundary and next experiment

The accompanying adapter consumes a supplied region of interest from an original unfiltered RGB frame, resizes it, extracts features, and reports the class and an uncalibrated margin gap. It does

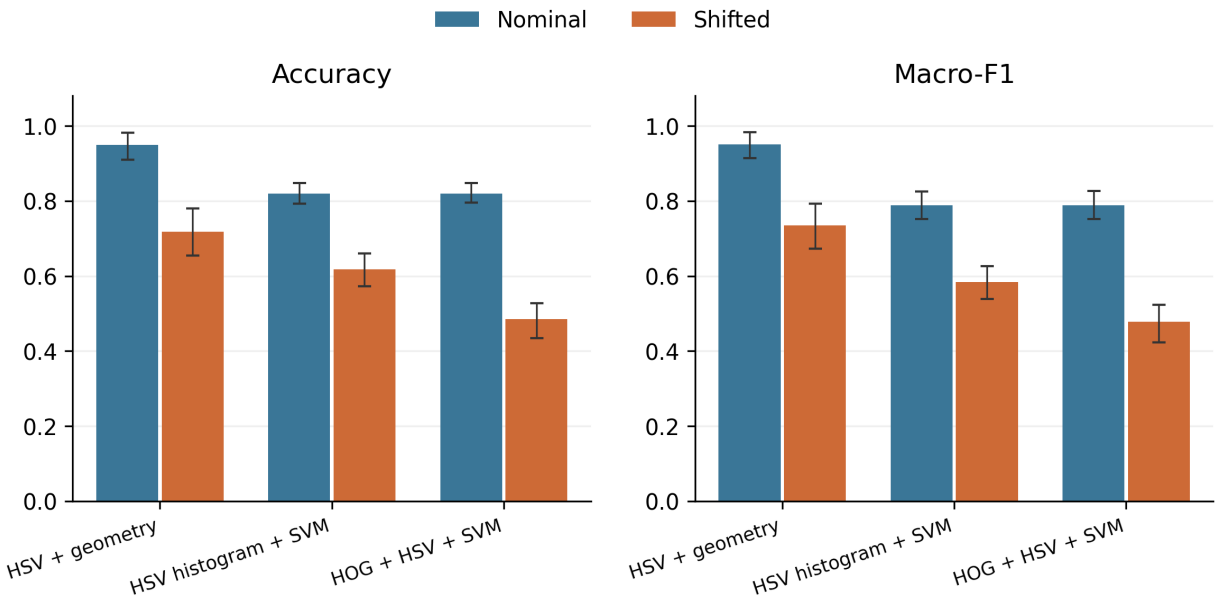


Figure 21: Executed classification results under nominal and shifted synthetic distributions. Error bars use whole-scene bootstrap resampling. The explicit baseline performs better than both learned models; the graph is not a field-performance comparison.

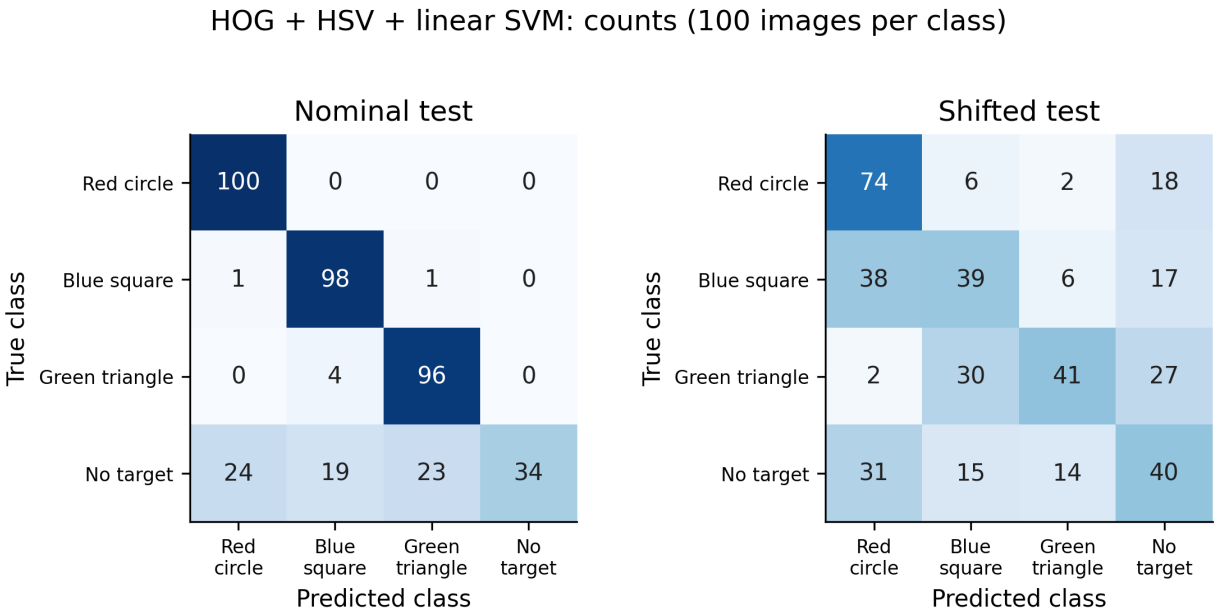


Figure 22: Confusion matrices from the executed synthetic benchmark. Counts correspond to 100 images per true class in each test set. The no-target column/row behavior is essential when assessing whether classification could assist board association.

not autonomously locate the board, produce a verified confidence probability, issue NASA commands, or control the motors. Classification is therefore a proposed aid to target labeling and later board association, not a demonstrated dependency of the mandatory command path.

Fixed target color–shape associations mean that the experiment does not establish color-independent shape recognition. Mismatched negatives reduce the shortcut but remain a finite generator-defined mixture. Real printed markers introduce reflectance, weathering, lens response, automatic exposure, occlusion, motion, and clutter that the present generator does not validate. Resizing an ideal supplied crop also avoids the practical problem of detecting and cropping a small board in a full frame.

The next useful experiment is a real-camera dataset with separate acquisition sessions for training and testing, including wrong-color shapes, blank boards, background clutter, partial views, multiple ranges, and direct sunlight or shade. Report localization errors separately from crop-classification errors. If a rejection threshold is introduced, fit it without using the final test set and report false acceptance as well as missed targets. Only target-hardware measurements can establish latency within the command budget. Until those checks are complete, the geometric baseline is the more defensible comparison point and ML remains an evaluated research extension.

10. Integrated Verification, Cost, and Research Limitations

10.1. Verification matrix

The analytical and software work identifies the measurements that carry the most decision value. A measured full-charge speed determines limiter settings. A sustained loaded-motor test addresses the climb-torque question. Corner weights and suspension travel identify the actual support geometry. Voltage traces under simultaneous loads address resets and data loss. A camera–LiDAR board test addresses whether a valid-looking number belongs to the requested target. These measurements should be linked to a fixed hardware and software revision so that later modifications do not inherit evidence from an obsolete configuration.

Table 10: Integrated verification activities and present evidence.

Subsystem	Verification activity	Record needed	Current status
Envelope and mass	Measure fully equipped rover, including antenna	Dimension sheet and scale record	CAD dimensions and mass estimate only
Propulsion	Full-charge speed, sustained climb, repeated turns, stop/failsafe	Speed, current, temperature, stop behavior	Analytical scenarios only
Suspension	Loaded corner force–travel and articulation checks	Motion ratios, end stops, corner weights	CAD concept and sensitivity model
Printed wheels	Coupon, hub, tire/interface and repeated-load tests	Process settings, forces, displacement, damage	Literature-informed test plan
Task power	Concurrent camera, display and servo operation	Rail voltage, pack current, temperature	Energy/current scenarios only
RF reception	Complete sequence with motors and converters operating	Received packets, errors, queue timestamps	Architecture and adapter contract
Imaging	Repeated commands, filter replacement, timestamps and storage	Fresh images and command-linked records	Mocked software checks
Ranging	Known distances, board angles, lighting and wrong surfaces	Raw frames, calibration, reference definition	Geometric analysis and adapter contract
Vision	Session-separated real-camera dataset	Localization, confusion matrices, rejection metrics	Synthetic experiment completed

Subsystem	Verification activity	Record needed	Current status
Integrated excursion	Full configured mission and immediate onboard review	Versioned event log and results	Not yet performed

The table avoids assigning an invented acceptance tolerance. Where the handbook specifies a condition, that condition governs. Where a system-level tolerance is absent or ambiguous, it must be established before the test. A pass/fail statement without a defined quantity, reference, operating condition, and uncertainty is not useful evidence.

10.2. Build budget and design-change control

The supplied component budgets list BDT 72,240.00 for chassis and drive items and BDT 110,854.30 for task items, giving BDT 183,094.30. These are planning costs, not purchase receipts or a complete lifecycle estimate. They exclude unrelated travel and event expenses. Different earlier worksheets used different currency conversions; reporting the technical component budget in BDT avoids combining inconsistent exchange-rate bases.

A design change should update the component list, mass estimate, wiring diagram, analysis inputs, and test record together. For example, replacing a motor changes speed and current assumptions; changing a wheel changes speed, torque leverage, clearance, and print validation; moving a battery changes load distribution and center of mass; replacing a camera lens changes pixel geometry and calibration. This controlled relationship is more valuable than adding unsupported precision to one table.

The estimated mass is the clearest cross-disciplinary tradeoff. Removing 1.847 kg would reach the 20 lb scoring objective, but thinning a printed rim or shortening a bearing span without load verification could compromise reliability. The appropriate sequence is to identify nonstructural mass, quantify the consequence of each change, and then recheck strength and operation. The present study does not prescribe an unvalidated lightening pattern.

10.3. What the study establishes

The work establishes mathematical consistency for the declared geometry and scenario parameters, executed behavior for the reference software with mocks, and reproducible performance for three classifiers on the declared synthetic crops. It also provides four editable Blender scenes and six editable Canva architecture diagrams for communicating the design. Those visual artifacts explain the proposed mechanism and task flow; they are not multibody-dynamics results, finite-element solutions, or records of task completion.

The strongest limitations are the absence of native CAD, measured component performance, physical camera data, calibrated center of mass, and full-system trials. The analytical model is quasi-static and uses idealized contact and optical geometry. Printed-material properties are not characterized. The supplied motor class lacks sufficient continuous torque/current information for a completed drivetrain selection. The servo's total usable pan range and the RC distance reference still need resolution. The synthetic vision study has no simulation-to-real validation and reports no benefit from the added HOG model.

These limits define a practical research path. A subsequent experimental paper should add measured methods, uncertainty budgets, repeated trials, and failure cases rather than rename the current calculations as experimental results. Literature supplies methods and comparison context; it cannot fill missing observations for this particular rover.

11. Reproducibility and Artifact Provenance

The accompanying source package contains the canonical manuscript, IEEE-style bibliography, numerical scripts and CSVs, reference executor, synthetic generator, trained classifiers, predictions, figure sources, Blender project, and Canva editing links. Analytical scripts regenerate six multipanel figure groups and verify unit identities, wheel-dimension closure, mass conversion, and equivalent force–velocity versus torque–speed power. The software suite regenerates its mock-adapter results.

The ML verification reloads saved models, reproduces their test predictions, recomputes metrics, and checks selected generated-image hashes.

```
python research/analysis/rover_analysis.py
python research/software/check_executor.py
cd research/ml
python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python vision_experiment.py --output reproduced
```

The supplied CAD screenshots are preserved as design evidence. New Blender scenes are explicitly procedural approximations from those views, with embedded parameters and source code. The Blender task scenes and the synthetic classifier dataset are separate products: no classifier result is attributed to the task-scene renders. Canva block diagrams are retained as editable designs; their rendered figure provenance is recorded with the source package. Published comparison figures are limited to the two identified CC BY 4.0 reproductions, with original figure numbers and attribution in their captions. Copyrighted full-text downloads are not redistributed as part of the package.

The literature log preserves the 20 queries, metadata deduplication, selected references, access levels, and claim limits. Numerical and artifact checksums support reproducibility and version identification; they are not certificates of physical correctness. All calculated, software-tested, and proposed elements remain distinguishable in the manuscript and data files.

12. Conclusion

This study presents a traceable design and software feasibility assessment of a four-wheel rover with commercial differential drive and autonomous execution of received imaging and ranging commands. The latest supplied geometry supports consistent dimensional calculations, while the estimated 10.919 kg mass exceeds the 20 lb scoring objective by 1.847 kg. A nominal 200 RPM wheel speed implies 8.105 km/h, showing that the motor label and an assumed loaded-speed reduction are insufficient to establish the speed limit. The calculated 20° climb case requires at least 1.255 N · m per wheel under the declared resistance assumption, before turning, transient and other unmodeled demands.

The sensing analysis shows that nominal image sampling is only one part of task feasibility. At 50 ft, a 1° aiming error shifts the ranging ray by approximately 266 mm, making target association, calibration, and the distance reference central integration requirements. Separate battery and regulator branches, correct low-side return routing, bounded command execution, fresh data, and visible failure records provide a coherent implementation basis. Thirty mock-adapter checks verify reference-software behavior without claiming hardware execution.

The synthetic experiment supplies a reproducible negative comparison: HSV-and-geometry rules outperform both linear SVM models, and adding HOG reduces shifted accuracy rather than improving it. Learning-assisted recognition therefore remains an extension requiring real-camera evaluation. Together, the calculations, code, and visual models identify a concrete route from the supplied design to a testable prototype. Physical mobility, electrical, sensing, and complete-sequence measurements are the remaining evidence needed to establish actual rover performance.

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